HRMA

Rocket Motor Analysis & Design Suite

Hybrid • Solid • Liquid Rocket Motors

User Guide

Install it, run a motor, read the verdicts

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Scope. HRMA is a preliminary-design and educational tool built on closed-form and one-dimensional engineering correlations. It is *not* a flight-qualification tool. Predicted performance should be cross-checked against an independent code (CEA / RPA / openMotor) and verified by physical testing before firing any motor. Section 13, "Scope and Limits", states what has been measured and what has not.

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1. What is HRMA?

HRMA (Rocket Motor Analysis Suite) is a desktop application for the preliminary design and analysis of hybrid, solid, and liquid rocket motors. It runs a calculation engine on your own machine, inside its own application window, and presents the results in a dark interactive interface: a performance summary, a warnings panel, dimensioned design tables, an interactive 3D model built from the solver output, an Analysis Dock of engineering panels, and working CAD, drawing, and report exports.

1.1 What the program does

You give HRMA an operating point (thrust, burn time, chamber pressure, propellants) and the hardware choices that go with it (materials, wall thickness, nozzle type, injector type). HRMA sizes the motor, predicts its performance, draws it, and tells you where the design sits against the limits of its own models.

- **Three designer pages:** Hybrid, Solid, and Liquid, each with its own input form, results, and exports.
- **An Analysis Dock:** fourteen engineering panels (structural, thermal, flow, feed system, safety, uncertainty, validation, and more) that run on the motor you just calculated.
- **An Analysis Center:** a component-by-analysis matrix for the heavier solves (finite-element structural and thermal, plane-strain grain, nozzle CFD, chamber acoustics, combustion stability).
- **Exports:** STEP, STL, DXF, technical-drawing PDF, report PDF, Excel and CSV tables, and an OpenRocket-compatible `.eng` motor file.
- **A launch site page:** pick a point on the globe and fly the motor you calculated along a solved flight path.
- **A formula reference:** the relations the solvers work from, typeset and grouped into fourteen sections.

1.2 Who it is for

Rocket teams, undergraduate and graduate students, faculty, and anyone who wants to follow motor physics end to end. The program does not design a motor *for* you: it turns the parameters you enter into geometry, performance numbers, and warnings, which you then judge with engineering reasoning.

1.3 Everything runs on your computer

The calculation engine, the propellant and material databases, the chart library, and the 3D viewer are all bundled with the application. No design data is uploaded anywhere, and the program works with the network cable unplugged. There are exactly two places where HRMA can reach the internet, and you control both:

- **The update check at startup**, which asks GitHub whether a newer version exists. If it fails, the program continues normally, and you can switch the check off in Settings.
- **The *Use online enrichment (Open-Meteo)* checkbox** on the Launch Site page, which is off unless you tick it. With it off, the site is resolved from local data only.

1.4 The rule the whole program follows: fail closed

HRMA does not invent a number it cannot compute. This is the single habit that explains most of what you will see on screen:

- If an input is missing, the program does not substitute a plausible value silently. Either it refuses the request and names the field, or it applies a canonical default *and declares it by name* in the "assumed on your behalf" block.
- If a model cannot cover your case, the result says so instead of producing a number. You will see `NOT MODELLED`, "no verdict published", `NOT EVALUATED`, or "no published data" where an answer would have been guesswork.
- If a verdict cannot be earned, it is withheld even when the underlying number is still shown. A drawing is not produced from a dimension the program had to assume.
- Every published value carries its source: `COMPUTED`, `USER`, `ASSUMED`, or `NOT MODELLED`.

Read the gaps as information. An empty field with a reason next to it is the program telling you that your design is outside what it can honestly cover. That is more useful than a number you cannot trace.

1.5 How to use this guide

Sections 2 and 4 get you installed and oriented. Section 5 works through one complete example motor, a hybrid burning HTPB with liquid N_2O at 1000 N for 10 s, and the steps generalize to the other two motor types.

Sections 6 and 7 are the engineering core of this guide. Section 6 takes a mission statement and turns it into a motor, decision by decision, naming at each step the field you type into, the number the program publishes back, and the band that number has to land in. Section 7 is the other half: how to read what comes back, which of the three performance numbers blames which piece of hardware, how far the models have been measured to be off, and how to tell an earned verdict from a read-back.

Sections 8 to 10 cover the analysis panels, the motor-type differences, and the exports. Sections 11 and 12 are the troubleshooting pair: what the program refuses and why, then the field guide to what you actually see on screen when it does. Section 13 is the one to read before you trust a number: what the models have actually been validated against. Appendix C is the field-by-field reference, with the accepted range for every input and what happens when you leave it empty.

All screenshots come from the application itself. Field names in this guide are written exactly as they appear on screen in the English interface.

2. Installation and First Launch

2.1 Installing

Each release ships two installers. Both embed a Python runtime and every library HRMA needs (thermochemistry, CAD kernel, fluid properties, charts), so you do not install Python or download anything at first run.

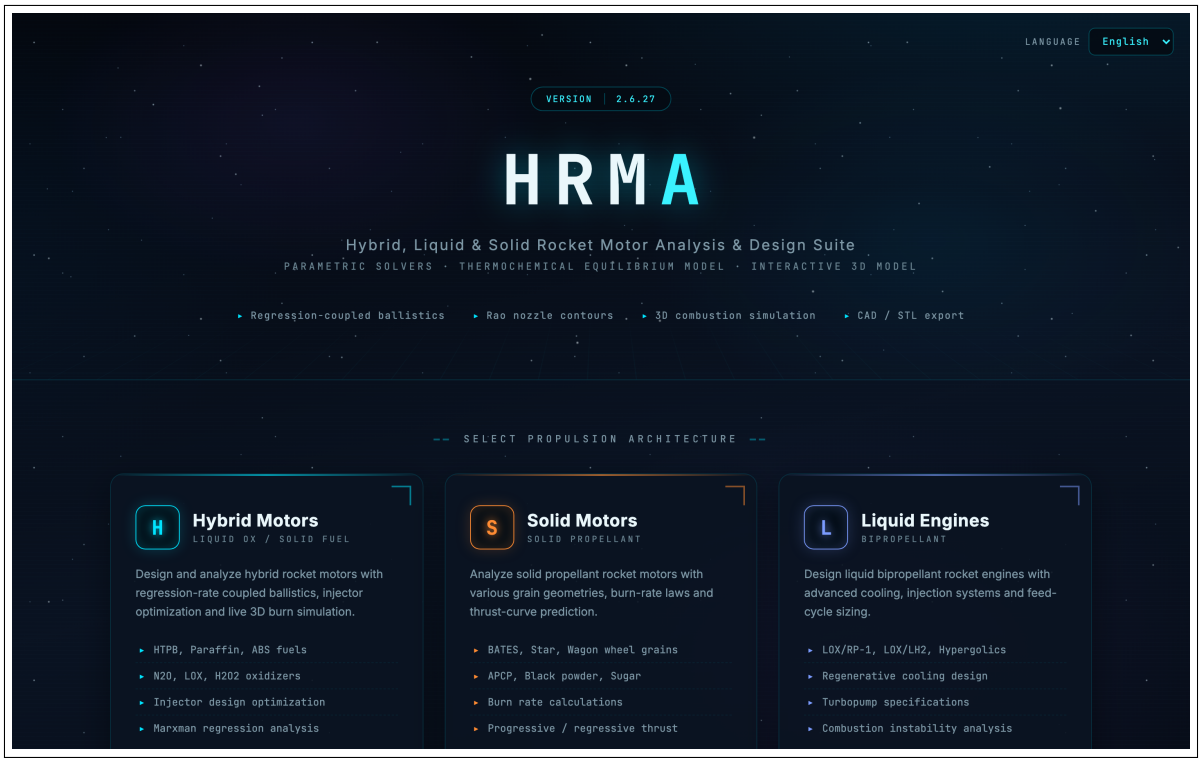
- **Windows 10/11:** download `HRMA-Setup-3.0.0.exe`, double-click it, and follow the wizard. The installation is per-user (it goes under your own `LOCALAPPDATA`), so no administrator rights are required. Windows SmartScreen may warn you because the installer is not code-signed: choose "More info", then "Run anyway".

- **macOS 11+ (Apple Silicon):** open `HRMA-Setup-3.0.0-macOS.dmg` and drag `HRMA` into `Applications`. The macOS build is for Apple Silicon and does not run on Intel Macs.

Unsigned packages. Neither installer carries a paid developer signature, so the operating system asks you to confirm the first launch (SmartScreen on Windows, Gatekeeper on macOS). The macOS application bundle itself is signed locally and the release process verifies that signature before publishing; an unsigned bundle stops the release.

2.2 First launch

HRMA opens its window immediately and shows a loading page while the calculation engines load in the background; the home page replaces it as soon as the engine answers. You get three motor-type cards, a version badge, and the links in the top bar.



The home page: cards for Hybrid Motors, Solid Motors and Liquid Engines, the Formula Reference link, and the top-bar controls (Language, Release Notes, User Guide, Settings).

If HRMA is already running, starting it again does not open a second copy; the existing window is used. The application picks a free local port for itself, so nothing needs to be configured.

2.3 The top bar and the window menu

Four controls follow you from page to page:

Control	What it does
Language	Switches the entire interface between `English` and `Türkçe`, chart labels included. The Settings panel keeps the same setting in sync.
Release Notes	Opens the release notes for the installed version in a panel.
User Guide	Opens this guide as a PDF in your system PDF viewer, in the language the interface is currently set to.
Settings	Opens the settings panel (language, updates, version information, bug report).

The same actions, plus *Check for Updates…* and *Open Output Folder*, are available from the native window menu, which is where to look if the page has not finished loading. Appendix A lists the menu items and where each one sits on macOS and on Windows.

2.4 Where your files go

Everything HRMA writes for you goes into a single folder, `Documents/HRMA`, with exported CAD files under `Documents/HRMA/cad_exports`. *Open Output Folder* in the window menu takes you straight there. Saved projects, exported drawings, reports, and the application log all live under that folder, so backing up one directory backs up your work.

3. What Changed in v3.0

This section describes the changes a user coming from 2.6.26 will notice first on screen in v3.0. (The work announced in the application as "2.6.27" ships with this release; no separate 2.6.27 was published.) Every screenshot here was taken from the v3.0 application itself, with the example projects; none has been edited. The concepts are explained in more depth elsewhere: reading the results in Section 7, the Analysis Dock in Section 8, exports in Section 10.

3.1 Outside the model validity limit: "verdict withheld"

The structural model evaluates the wall at a given temperature. If that temperature reaches the melting point of the material, or goes past the last point of its strength-derating data, the strength is no longer known. In that case HRMA does two things:

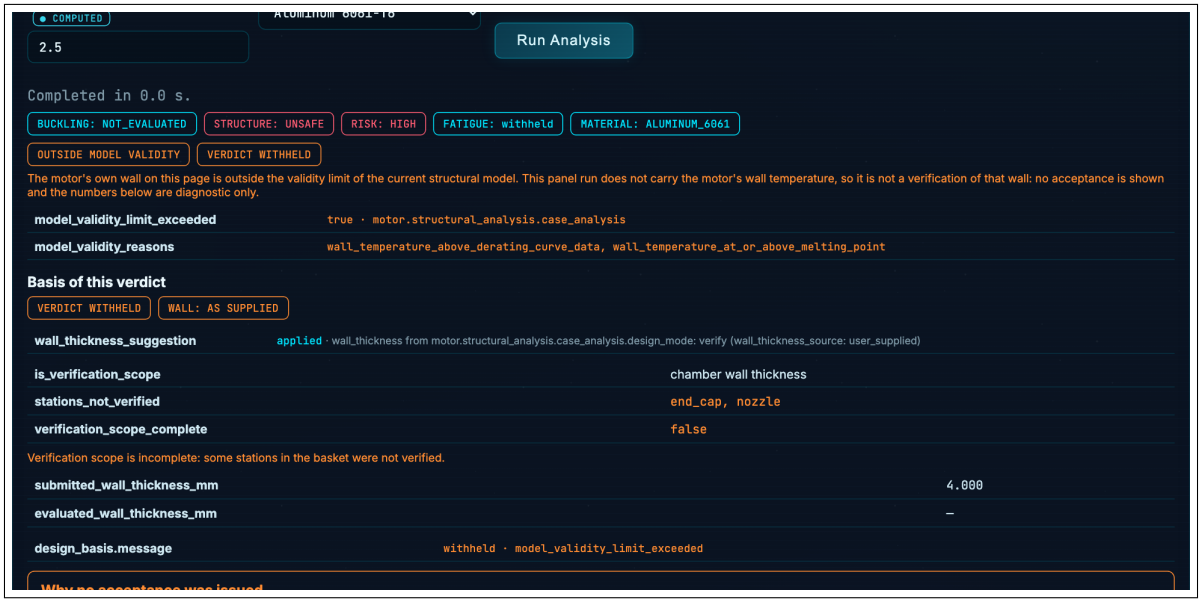
- It shows the **OUTSIDE MODEL VALIDITY** badge and writes the reason codes:
  - `wall_temperature_at_or_above_melting_point`: the wall is at or above its melting point;
  - `wall_temperature_above_derating_curve_data`: the wall is above its strength-derating data.
- It **withholds** the acceptance and the numeric design verdict. The safety-factor row shows **VERDICT WITHHELD** instead of a number; no PASS, SAFE or "verified" acceptance is shown.

Stresses, safety factors, wall thickness and masses stay on screen, but they are *diagnostic* only: they are not valid design outputs. A FAIL, UNSAFE or MARGINAL status from the current model is never suppressed; it stays visible as a warning.



Solid motor, Structural Analysis tab: aluminium case, no insulation, long end-burner grain. The case heat model finds the case above its melting point; the safety-factor row reads "VERDICT WITHHELD" and two reason codes are listed.

The same rule applies on every surface that carries a verdict: the result cards, the end closures, the acceptance sentences of the sizing mode and the structural panel of the Dock. The Dock panel reads the motor's own validity flag and states where the flag came from, for example `model_validity_limit_exceeded true · motor.structural_analysis.case_analysis`.



Analysis Dock, STRUCTURAL tab, same motor: the "OUTSIDE MODEL VALIDITY" and "VERDICT WITHHELD" badges, the source of the flag and the "Basis of this verdict" block. The STRUCTURE: UNSAFE status stays visible.

What this means, and what it does not. When the wall temperature comes from the heat-transfer module it is a *bare-wall* steady-state estimate in which protection by the grain, liner or insulation is not modelled. An outside-validity result is a limit of the current model; it is not a statement that a real motor of this design is impossible. To bring the result inside the validity limit, revisit the material, the insulation, the cooling or the wall temperature. For motors inside the validity limit, verdicts and acceptance sentences appear as usual.

The safety panel follows the same principle. For propellant classes without a sourced TNT

equivalent (hybrid, sugar propellant, black powder) the TNT equivalent, blast distance, fragment range and evacuation distance are shown as **NOT_EVALUATED**, with the reason named. No number is invented for these classes any more.

3.2 Thermal results: allowable temperature from the material record

The thermal panel of the Dock plots the wall temperatures together with two limits of the material: the *allowable temperature* and the *melting point*. Both come from the record of the selected material. The note limit on the heat-flux chart also comes from the allowable step. In the screenshot below the limits for Aluminium 6061 are 477 K allowable and 855 K melting; the wall is far above both, so the thermal risk is HIGH and the "WALL T UNPHYSICAL — cooling insufficient" badge is red.

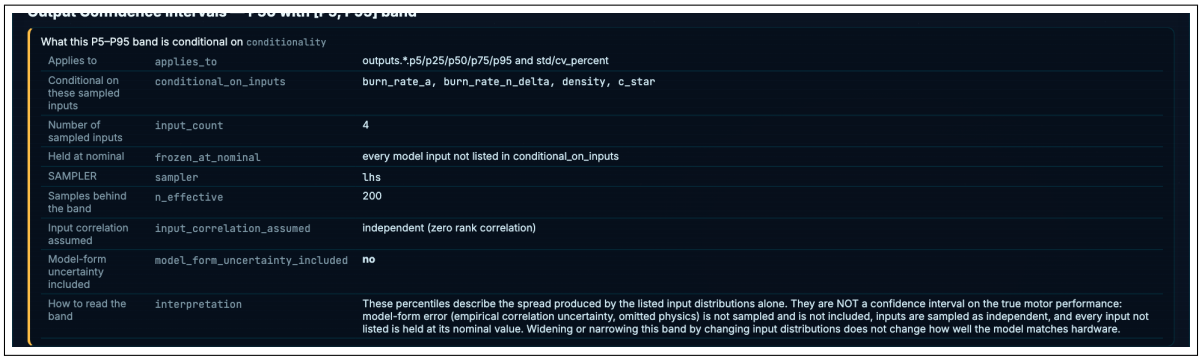


Analysis Dock, THERMAL tab, same solid motor: risk and cooling badges, wall-temperature cards and the "Wall Temperatures vs Material Limits" chart.

The "Heat Load & Cooling Assessment" table under the panel states the basis next to each number. For example, when a value rests on a generic default exit diameter this is shown next to the row with a `generic_defaults_used` note.

3.3 Panel provenance notes

In v3.0 the panels of the Dock and the Analysis Center write *what a result rests on* next to the result. These lines are provenance notes. They tell you which field a value was read from (its source path), whether a default was applied, which inputs the result is conditional on, and what was not measured.



UNCERTAINTY panel, example solid motor: the "What this P5–P95 band is conditional on" table. The band comes only from the distributions of the listed inputs; model-form error is not sampled, and the band is not a confidence interval on true motor performance.

Other examples:

- **Stability panel (chug):** the sensitivity to the choice of the time lag τ is shown under its own heading. When the result depends on the choice of τ, the frequency and growth-rate bands across the τ candidates are given; when it was not measured the panel says "NOT MEASURED", which is not the same as "insensitive". A source line names the block the value was read from.
- **Transient panel (hybrid blowdown):** when the burn is cut off, the oxidizer left in the tank and the impulse lost at cut-off appear as badges together with their field names. If a value is not published, the badge is not shown at all; it is never printed as "0".
- Panel texts, badges, table headings and chart axes follow the selected interface language. Machine field names (e.g. `model_validity_reasons`) stay the same in both languages so that they can be searched for in a bug report.

3.4 Hybrid warnings

The warnings list on the hybrid page grew in v3.0. For the example hybrid project (N_2O/HTPB) you see the following after Calculate:



Warning list of the example hybrid motor: injector correlation range and pressure drop, the measured bias of the HTPB regression coefficients, throat and wall temperature (CRITICAL), fields the model solves itself, cavitation and flash-boiling risk.

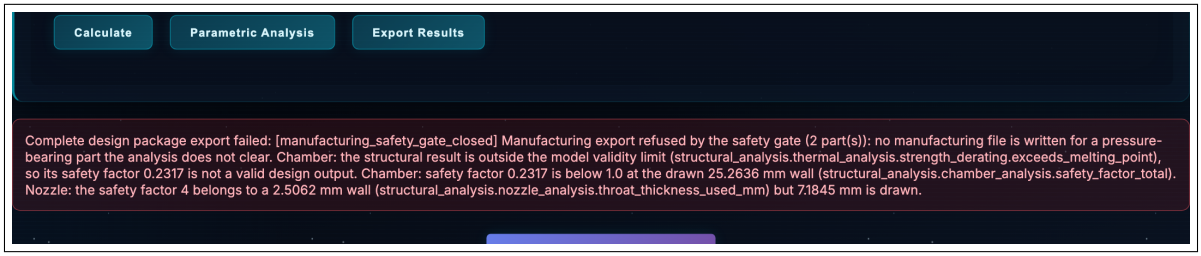
- **N_2O inlet state:** the tank temperature and phase of the oxidizer are now inputs; the default is saturated liquid at 293.15 K. The vapour pressure P_v in the cavitation warning is computed from this temperature.
- **Tank pressure in self-pressurised N_2O mode:** the injector circuit takes its feed pressure from the saturation pressure at the tank temperature, $P_{sat}(T_{tank})$. If the tank pressure you enter differs from it by more than the HRMA tolerance, a "Tank pressure … was not used" warning appears; to change the feed pressure, change the oxidizer temperature. This is why the example project carries its tank pressure at P_{sat}.
- **Injector:** a Reynolds number outside the validity range of the correlation and a pressure drop that is too large relative to chamber pressure are reported separately.
- **Regression coefficients:** the measured bias of the built-in HTPB coefficients against the validation database is written out campaign by campaign. The coefficients are deliberately not re-tuned; you can enter your own a and n from your own firing data.
- **Fields the model solves:** for a field that is not read from the form but solved by the model (e.g. the expansion ratio) the warning states the value the analysis used.

3.5 CAD export gate and refusal messages

If a pressure-bearing part (chamber, nozzle, tanks) is not cleared by the analysis, HRMA **does not write its manufacturing file**. The manufacturing STEP package, the DXF manufacturing drawing of the same package, the complete design package and the tank CAD package are refused on the server; no file reaches the browser. The gate closes when:

- the safety factor is below 1.0;
- the safety factor is not published or is not a finite number;

- the structural result is outside the model validity limit;
- the safety factor belongs to a wall thickness other than the drawn one.



Example hybrid motor, "Generate All Files": the manufacturing safety gate refuses the package for 2 parts. Each reason is written with the part name, the number and the field the number was read from.

The refusal message is not a raw server response. It starts with the machine code, then a summary sentence, then the reasons part by part. The machine code for this gate is:

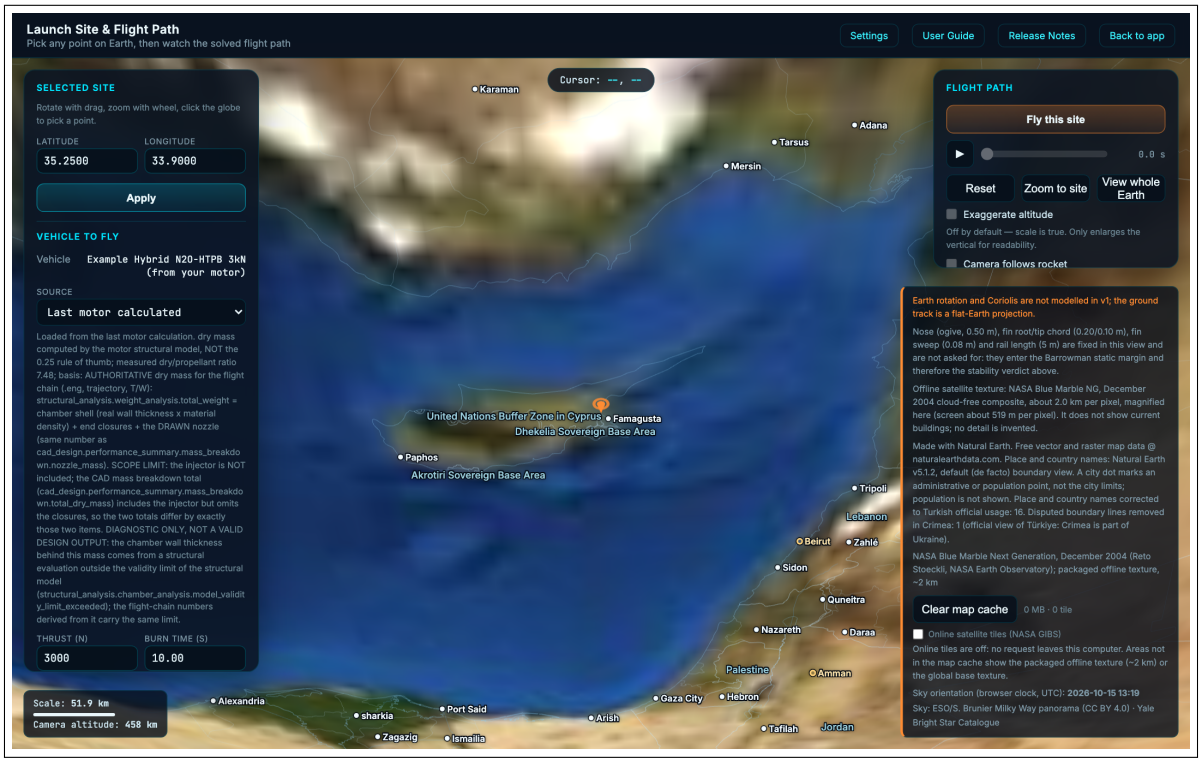
`manufacturing_safety_gate_closed`

The message does not echo back the text you sent.

There is a separate gate for the grain section. The CAD generators can draw only a single circular port. A star or finocyl grain, or a multi-port grain, is therefore not drawn as a plain tube; the export is refused with a "fuel grain section cannot be drawn" reason. The tank CAD follows the flat-ended tank model used by the solver. The nozzle STEP is a smooth surface of revolution through the contour points; no conical facets show in the CAD program.

3.6 Launch Site map: detail, Turkish names, privacy switch

The globe on the Launch Site page gained offline vector detail in v3.0: coastlines, country and province borders, lakes and rivers, city and country labels. As you zoom in, a more detailed level (Natural Earth data at 1:110, 1:50 and 1:10 million) is loaded. This data ships with the application; no internet is needed.



Launch Site, around Cyprus (camera altitude 458 km): the packaged offline Earth imagery, place names ("Dhekelia Sovereign Base Area", "Akrotiri Sovereign Base Area", "Famagusta"), provenance notes at the bottom right, the "0 MB · 0 tile" cache counter and the "Online satellite tiles" switch, off.

Turkish official names. Place and country names are corrected to Turkish official usage. In the Turkish interface this gives, for example, Güney Kıbrıs Rum Yönetimi, Dikelya, Elazığ, Akmescit, Gözleve, Hankendi, İdlib, Bükreş and Temeşvar; in the English interface the country label reads "Türkiye". The disputed boundary line in Crimea has been removed; Crimean settlements are shown as part of Ukraine. The provenance note at the bottom right states how many names were corrected and how many lines were removed.

City dots. A city dot marks an administrative or population point, not the city limits. Lines are generalised data; at close range their positions can be off by hundreds of metres up to kilometres.

Privacy switch. Satellite tiles come from NASA GIBS. Every tile request tells the provider which area you are looking at. The "Online satellite tiles (NASA GIBS)" switch is therefore *off by default*:

- **When off,** no request leaves the computer. The packaged NASA Blue Marble texture that ships with the application (December 2004, about 2 km per pixel at the equator) is drawn without any network; areas not in the map cache show this texture or the global base texture. In the session of the screenshot the packaged tiles came from the local server (HTTP 200), the other tile requests were refused locally (HTTP 403) and no request reached the outside network.
- **When you switch it on,** a warning text appears and tiles are requested below about 1,500 km altitude. The state of the switch is stored in this browser; if it cannot be read, it is treated as off.
- The "Clear map cache" button deletes downloaded tiles; the counter next to it shows the cache size.

3.7 Other changes you will notice on screen

The changes below went in just before v3.0 was released. The full list is in the in-app Release Notes window and in `CHANGELOG.md`.

- **The regenerative cooling panel no longer invents values.** The panel used to pre-fill values that had no source: channel count, channel width and depth, liner thickness, coolant, inlet temperature and inlet pressure. Now every field either takes the value the engine publishes or stays **empty and required**. No request is sent while a required field is empty. In practice this means: if the engine does not publish the channel dimensions, *you have to enter* the channel count, channel width and depth. On the API, a `/api/regen-cooling` request without `n_channels` gets a named 400 error. When the coolant station march cannot run, the thermal verdict does not silently disappear; a warning says so.
- **No English left-overs and no point decimals in Turkish mode.** Fixed places: the Structural tab of the solid motor (risk classes, closure bolt note), the structural and thermal panels of the Analysis Dock, the vehicle provenance and upload warnings on the Launch Site, the conditions of the uncertainty panel and the material names in hybrid warnings. Percentages are written in Turkish form (for example %16,2).
- **The safety panel says why the thermal axis was not evaluated.** In an uncooled hybrid the heat module marks its own equilibrium wall temperature as non-physical. The panel does not give a thermal verdict from that temperature; the thermal verdict and the combined

pressure-plus-thermal safety factor are withheld with the reason. The panel now states that reason; before, the case looked like a missing input.

- **The explosive equivalent follows the panel's rule in the API and the PDF report too.** For classes without a sourced TNT equivalent (hybrid, sugar propellants such as KNDX, black powder) the TNT equivalent and the distances derived from it become "not evaluated" in the API response and in the PDF report as well. The solid propellant class is taken from the catalogue family, so KNDX is no longer counted as a composite propellant. APCP and liquid results are unchanged.
- **The uncertainty analysis runs with a zero diameter input.** A zero diameter means "automatic", as in the engine calculation. The uncertainty run of the example hybrid engine used to fail with `non_physical_uncertainty_input`; it now runs.
- **Broken export data gets a named refusal.** When an inner block of the export payload (for example `design_summary` or `structural_analysis`) is not in the expected form, the export endpoints no longer return a server error (HTTP 500). Instead they return a 422 `invalid_export_geometry` that names the full path of the field. The STL package returns the assembly builder's named refusal (422) when the throat or exit diameter is not published. The manufacturing gate described above covers, besides STEP and DXF, the drawing PDF, the STL package, the single STL, the complete package and the `/api/export-cad` endpoint.
- **The CFD runtime note follows the selected mode.** A viscous run now quotes the worst-case runtime table for viscous runs; before, it quoted the value from the inviscid table. Both tables were re-measured with a single protocol. The default resolution level measured 18.5 s without numba. The viscous table was measured on a loaded machine and is therefore a cautious upper bound; the note says so.

4. Main Pages

4.1 The page map

Page	Address	Purpose
Home	`/`	Motor-type selection, version badge, formula reference
Hybrid Motors	`/hybrid`	Full hybrid motor design (the most complete page)
Solid Motors	`/solid`	Propellant catalogue, grain geometry, ballistics
Liquid Engines	`/liquid`	Feed system, injector, cooling, engine cycle
Formulas	`/formulas`	The relations the solvers work from
Launch Site	`/launch-site`	Globe, site resolution, flight path

4.2 What the three designer pages have in common

Hybrid, Solid and Liquid are laid out the same way, top to bottom:

1. **A project bar** with the current project name, an unsaved changes indicator, *Save As*, *Discard changes*, and *Import from CAD (STEP)*.
2. **Database badges** for *NASA CEA* and *NIST WebBook*. They open reading "not checked yet" and turn green only once the real check answers. A badge that is still grey means the check has not come back, not that something is broken.
3. **The input form**, split into numbered sections.

4. **A Calculate button**, plus the analysis-type selector on the hybrid page (single point, parametric, trajectory).
5. **The results**: a Performance Summary, a Warnings panel, the design tables, the injector plan, the cross-section, and the charts.
6. **Standalone panels** for the injector, the transient solution, tank blowdown, and 6-DOF flight dynamics.
7. **The Analysis Dock** and, below it, the **Analysis Center** (Section 8).
8. **The 3D model** and the **Export & CAD Options** panel.

4.3 Projects: saving and reopening a design

The project bar stores the state of the form so you can come back to it. Use *Save As* to name a project (the program rejects names the operating system cannot use, including the Windows device names `CON`, `PRN`, `AUX`, `NUL`, `COM1`–`COM9` and `LPT1`–`LPT9`), and *Overwrite* to replace an existing one. While you have unsaved edits the bar shows *Unsaved changes*, and *Discard changes* throws them away.

When you reopen a project, HRMA tells you what it could not restore instead of restoring it approximately: fields saved in the file that have no matching input on the current page are reported as skipped, with a count, and a file it cannot read at all is listed as `CORRUPT` with the server's stated reason.

4.4 The Formulas page

Formula Reference on the home page opens the equation reference: fourteen sections covering performance parameters, nozzle design, combustion, solid, liquid and hybrid motors, heat transfer, structures, trajectory, safety, manufacturing, cost, and the relations behind the advanced-analysis figures. Sections 1–13 state the reference relations of the field; the last section ties those relations to the figures the advanced panels draw. Use it when you want to see which relation a result came from.

4.5 The Launch Site page

The Launch Site page opens a globe. Drag to rotate, scroll to zoom, and click to pick a point, or type a latitude and longitude and press *Apply*. *Resolve site* then reports the elevation, the terrain relief over the surrounding cells, the local gravity $g(\varphi, h)$ next to the standard g_0 that specific impulse is referenced to, and the local temperature.

You then choose what to fly: the last motor you calculated, a saved `.hrma` project, or an example vehicle. The vehicle fields (body outer diameter and length, airframe dry mass, C_{d0}, fins, fin span) sit next to the wind speed and direction, launch elevation and azimuth.

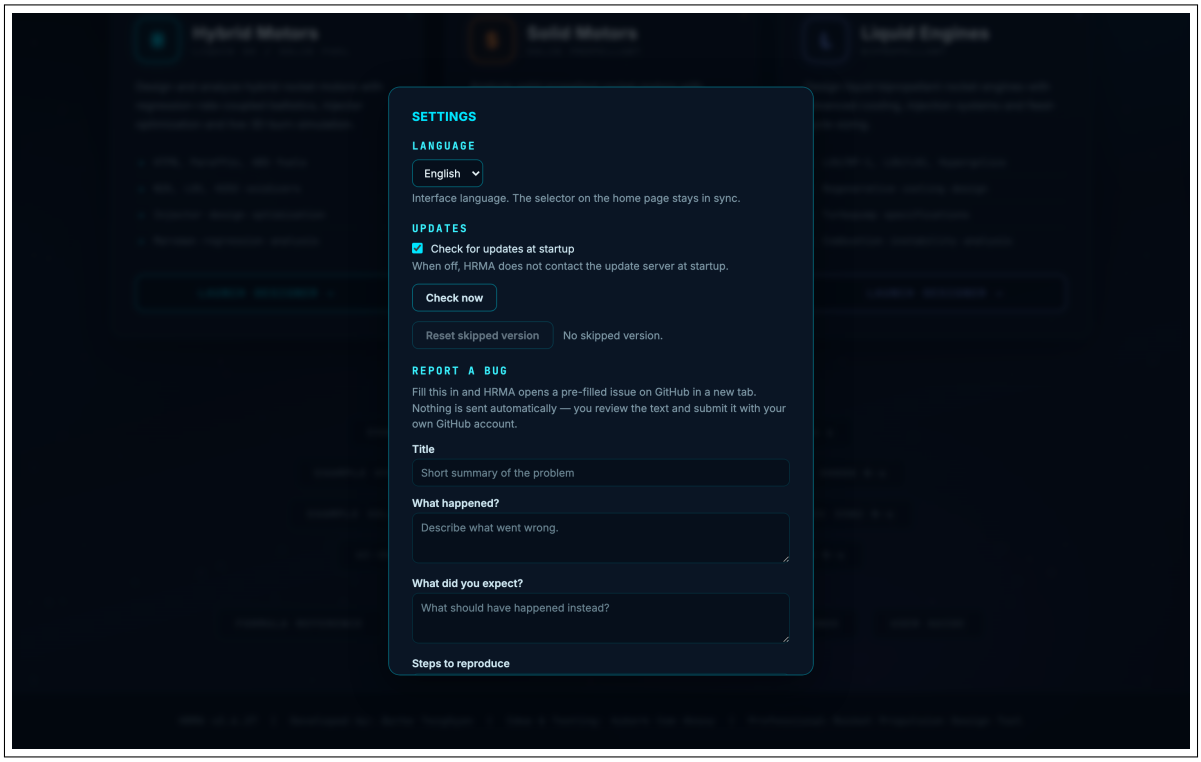
> **Two mass fields, two meanings.** *Airframe dry* mass *excludes* the engine inert mass shown above it; the solver adds them. Entering the total in both places flies a heavier rocket than you have.

4.6 Settings

The Settings panel gathers the application-level controls:

- **Language:** the same setting as the home-page selector.

- **Updates:** *Check for updates at startup* can be turned off, and when it is off HRMA does not contact the update server at startup. *Check now* runs the check on demand and reports either "You are up to date", a new version, or a failure to reach the server.
- **Skipped version:** if you once chose *Skip this version*, it is listed here, and *Reset skipped version* makes the program offer it again.
- **Application:** the installed version, and a link to HRMA on GitHub. If the version cannot be read, the panel says so and lists the sources it tried rather than printing a guess.
- **Report a bug:** a short form (title, what happened, what you expected, steps to reproduce). It opens a pre-filled issue on GitHub in a new tab; nothing is sent automatically, and you submit it yourself. The optional diagnostics attachment adds the numeric inputs on the page and the console errors captured after it loaded, with file paths, user names, e-mail addresses and launch-site coordinates removed. It is off unless you turn it on.



The Settings panel: language, the startup update check and the bug-report form. The panel scrolls inside itself; the skipped-version reset and the installed version sit further down.

5. Worked Example: Designing a Hybrid Motor

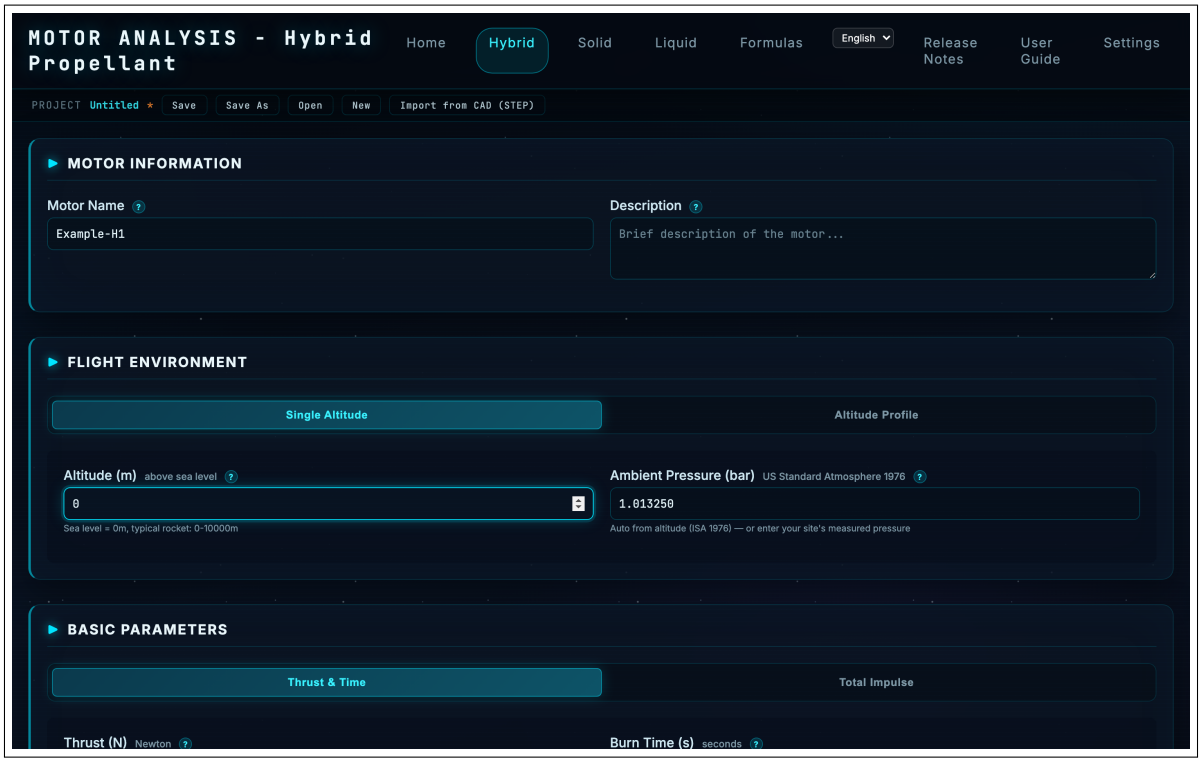
This section runs one motor from an empty form to an exported package. Open *Hybrid Motors* from the home page.

5.1 Step 1 — Fill in the basic parameters

The form arrives with the page defaults already in place. For this example we keep the propellant pair the page starts with and set the operating point:

Field	Value	Note
Motor Name	`Example-H1`	Used in report and export file-names
Altitude (m)	`0`	Ambient pressure follows the US Standard Atmosphere 1976
Thrust (N)	`1000`	Target thrust level
Burn Time (s)	`10`	Total impulse follows from thrust and time
Chamber Pressure (bar)	`20`	Absolute
Tank Pressure (bar)	`50`	Absolute; must stay above the chamber by the injector and line losses
Fuel Type	`HTPB`	Regression coefficients are filled from the database
Oxidizer Type	`Nitrous Oxide (N2O)`	Density follows the temperature you enter
O/F Ratio	`7.5`	The page default for the N_2O / HTPB pair
Nozzle Type	`Conical Nozzle`	Conical, bell or parabolic
Injector Type	`Showerhead`	Each type has its own parameters

The O/F default belongs to the propellant pair, not to your design. `7.5` is the tabulated theoretical optimum for N_2O with HTPB, which is the pair the two selectors start on. Choose a different fuel or oxidizer and the page tells you the default no longer matches: pick both first, then press *Find Optimum*, which sends the pair to the server and returns the optimum for it. No "distance from optimum" figure is printed, because the program does not know the optimum for an arbitrary pair until it computes it.



The top of the hybrid input form: Motor Information, Flight Environment and Basic Parameters. The

form continues below with the propellant, injector and design sections, where Find Optimum *sits next to the O/F field. Every field carries a "?" tooltip with a short explanation, typical values, and the effect of the parameter.*

Empty, zero, and automatic are three different things. This distinction matters more on this page than any other:

- **Empty** means "I am not sending this". An empty field is not transmitted at all, so the server can tell the difference between a value you left out and a value you deliberately set. Where the endpoint has a canonical default it applies it and lists it by name in the declaration block; where it has none, it refuses and tells you which field it needs.
- **Zero** is an explicit instruction on the fields that document it, such as *Initial Port Diameter (mm) 0 = auto*, *Ae/At Ratio expansion ratio, 0=auto* and *Chamber Diameter (mm) 0=auto*. There, `0` tells the solver to size the quantity itself.
- **A number** is used as given, after the band check described in Section 11.

The *Advanced Parameters* section holds the sizing levers: L* Value (m), Initial Port Diameter, Number of Ports, Ae/At Ratio, Nozzle Type, Combustion Type (infinite or finite area), Chamber Diameter, Contraction Ratio Ac/At, and Mass Flux. The *Fuel Configuration* section lets you override the database values for fuel density and the regression-rate coefficients a and n; do that whenever you have coefficients fitted from your own static fires.

5.2 Step 2 — Fill in the design configuration

The form above sets the operating point. *Configure Design* opens the *Design Configuration* panel, which says what the motor is made of and what margin it is sized to. Two tabs: *Motor Design* and *Injector Design*.

- **Chamber Material** (AISI 304 Stainless Steel, Aluminum 6061-T6, Inconel 718), **Wall Thickness (mm)** and the cooling selection feed the heat-transfer model (wall temperatures, heat flux) and the structural module, so the thermal and structural results describe the same motor. The first entry, AISI 304 stainless steel, is what the analysis uses unless you change it. Selecting cooling channels raises a warning: the coolant-side film coefficient is taken from the regenerative range in the literature, and coolant flow rate, channel velocity, pressure drop and boiling margin are not verified here. That is a declared limit of the model, not a fault.
- **Safety Factor:** the design factor the structural module aims at. Your real wall thickness is passed with it, so the module runs in verification mode and the reported minimum safety factor follows from pressure, diameter, material and thickness. A value outside the accepted range is refused with the range named, and a value that cannot be read at all is refused as not a number, in both cases without silently substituting a design value.
- **Chamber Length Override (mm):** overrides the chamber length derived from L*. Leave it empty for automatic. If your value is shorter than the fuel grain plus the pre-combustion chamber, the override is rejected and the warning quotes both your value and the minimum. It is not quietly trimmed, because a trimmed length would be a different motor from the one you asked for. A value above the accepted limit is rejected too, with the note that numbers that large usually mean a unit mix-up.
- **Nozzle Material** (graphite, tungsten, copper): drives the throat thermal check and the erosion estimate. The equilibrium wall temperature at the throat station is compared with the material's allowable temperature, and exceeding it is a critical warning that names both temperatures. Erosion is computed only for materials with a published coefficient band; tungsten has none, so no coefficient is invented and the result reads "no published data". The

copper option is labelled regeneratively cooled and is solved under that assumption. Erosion is reported, not fed back: the steady-state solution assumes a rigid throat, so use the transient solver for erosion-coupled thrust and pressure histories.

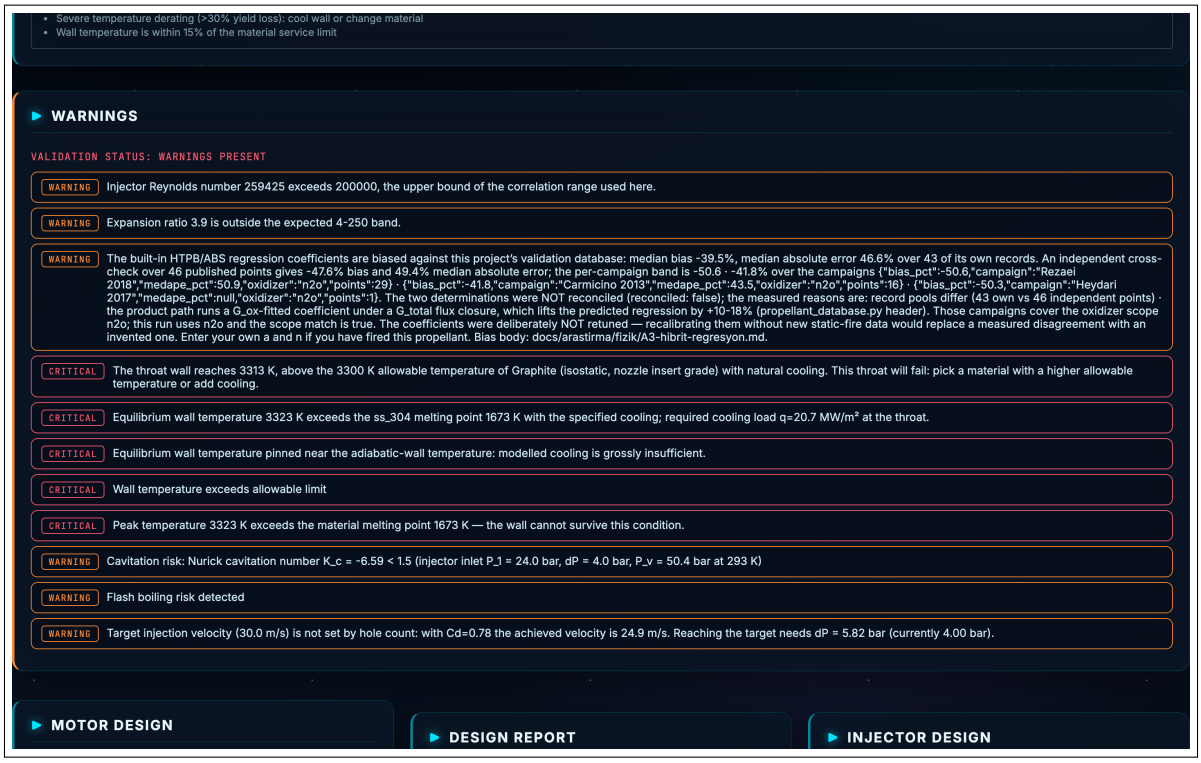
- **Injector Material** (AISI 316 Stainless Steel, Titanium Grade 5, Brass (Low Pressure)): sizes the injector plate. With the material's yield strength and density, the plate bending stress, safety factor, required thickness and mass are reported (edge-fixed circular plate, Roark's Formulas Table 11.2 case 10b; ASME BPVC PG-52 ligament efficiency for the drilled area).
- **Swirl injector fields:** if you select the *Swirl* injector, *Swirl Chamber Diameter* and *Tangential Entry Angle* enter the pressure-swirl solution: the chamber diameter is the D_s in $K = A_p/(D_s\, d_o)$, and the entry angle is taken as the target spray half-angle, from which the inverse solver picks K and sizes the exit orifice and the slots. Leave the slot width and height *empty* to design for a target angle. Fixing the slot geometry fixes the swirl number as well, so the target angle can no longer be honored, and the program says so rather than swallowing the input. Angles outside the solvable envelope are clamped with a warning, and a swirl chamber narrower than 1.25 times the exit orifice breaks the geometric assumption of the model, which is also reported.

5.3 Step 3 — Calculate

Press **Calculate**. The form is sent, the solver runs, and the results appear below. Four blocks matter:

Performance Summary. The headline numbers: thrust, total impulse, specific impulse, characteristic velocity c^*, thrust coefficient C_F, chamber pressure, throat and exit diameters, mass flows, and a safety-state badge. The badge text comes from the result, not from a fixed list, and nothing is printed before a calculation exists.

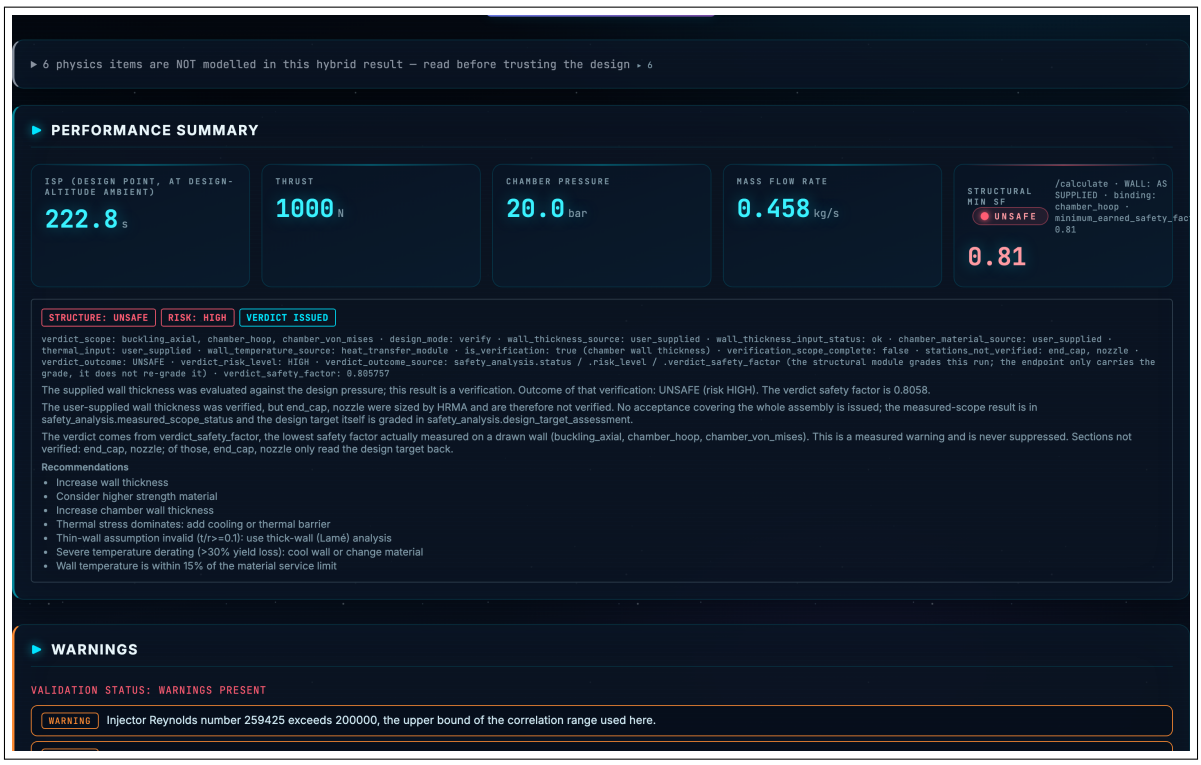
Warnings. Design-criteria messages from the solver and the validation layer. **Read this panel before you trust anything above it.** The messages are specific: a characteristic velocity outside the expected band for your propellant pair names the band; a low expansion ratio names the expected range; a chamber material that is not in the database names both what you asked for and what was used instead; flow separation in the nozzle states that thrust is reported on the separated effective expansion ratio rather than the geometric one. If the panel is empty, no warning was raised; it is not hidden when there is something to say.



The Warnings panel. Each message names the quantity, the band it is compared against, and what the solver did.

Design tables. *Motor Design* lists the full geometry (parameter, value, unit): nozzle contour dimensions and angles, grain and port geometry, chamber length with a statement of whether it came from L* or from your override, and wall thickness. *Injector Design* lists the orifice plan. *Design Report* summarizes the run.

Material verdicts. The throat temperature margin and erosion estimate for the nozzle material, and the plate stress, safety factor, required thickness and mass for the injector material. If a quantity cannot be produced, nothing is invented: the block reports that it was not analyzed and gives the reason. In particular, when the detailed injector circuit model cannot size the type you selected, the orifice plan (hole count, hole diameter, manifold diameter) is *not reported* rather than estimated, and the warning tells you not to take injector geometry from that run.



The Performance Summary, the Warnings panel and the design tables after a calculation. The values shown are the ones this example produced; your own run prints its own numbers.

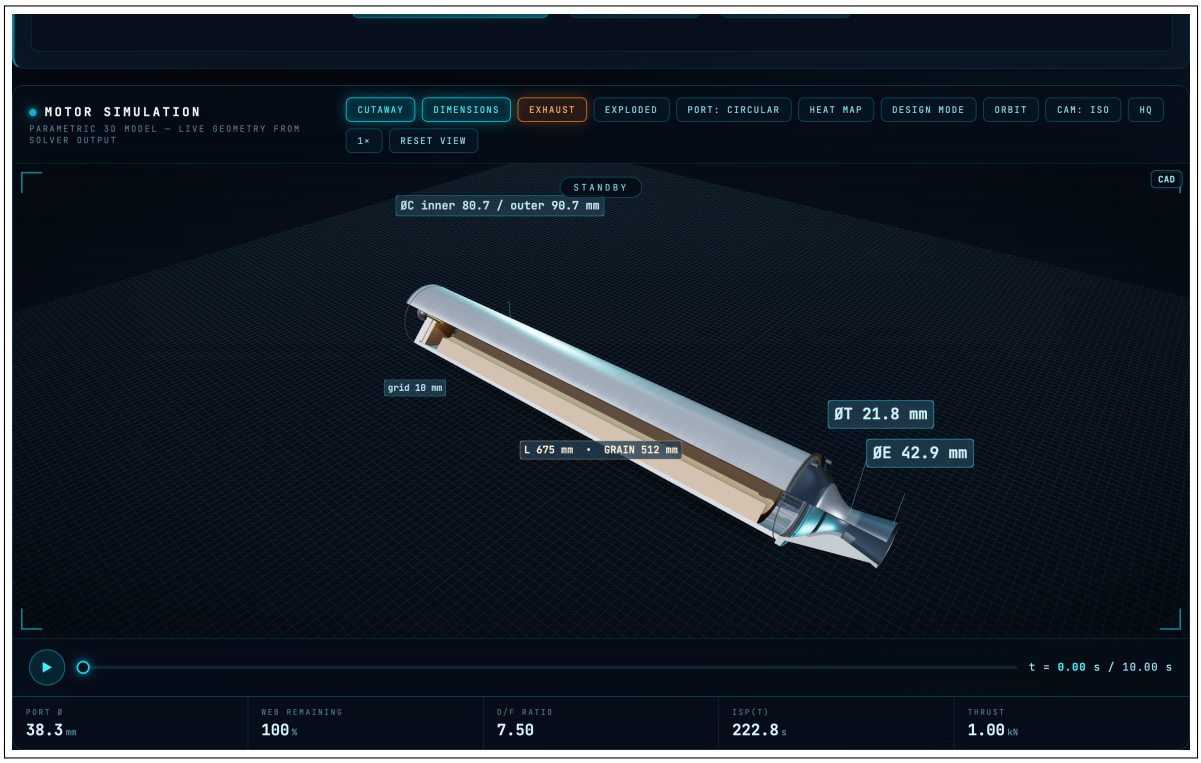
This example is deliberately not a "safe" design. With the values above the structural verdict comes out **UNSAFE** (minimum safety factor below 1), and the expansion ratio stays below 4 at the sea-level design point, which raises the geometry band warning. The injector orifice Reynolds number also comes out at 259,421, above the 200,000 upper edge of the correlation, which raises a second band warning. Five further messages carry the red CRITICAL badge: the graphite throat reaches 3313 K against a 3300 K allowable, and the chamber wall reaches 3323 K against the 1673 K melting point of AISI 304. The example is not thermally buildable as it stands; Section 6 works the structural and nozzle levers first. These are the places where the program stops you, and it does not hide them: changing the wall thickness, the chamber pressure and the expansion ratio to see how the verdict turns is the whole point of this guide.
The O/F ratio is worth a note of its own, because the band moved in this version. The page default of 7.5 is the tabulated optimum for the N_2O/HTPB pair, and it used to fall outside the validation band, so a user who opened the program and pressed *Calculate* without changing anything was told the mixture ratio was outside the usable range. The band was corrected rather than loosened: the upper edge moved from 6.0 to 12.0, which brings the optimum inside, and the lower edge moved from 0.5 to 4.0, which means heavily fuel-rich designs such as O/F 2.5 now raise a warning where they used to be silent. Section 6 explains why a storable oxidizer optimizes so high.

Warnings are the product, not a defect. Rather than invent a value for something it cannot compute, HRMA warns explicitly or says it did not model the case. When you see an orange or red message, cross-check that quantity by hand or with an independent code before you cut metal.

5.4 Step 4 — Inspect the 3D model

Below the results, *Generate 3D Model* builds an interactive model from the geometry the solver produced. The header reads *PARAMETRIC 3D MODEL — LIVE GEOMETRY FROM SOLVER OUTPUT*, and the toolbar gives you:

Button	What it shows
Cutaway	Sections the motor so you can see the port, the liner and the throat
Dimensions	Draws the dimension labels on the model
Exhaust	Adds the exhaust plume
Exploded	Separates the components along the axis
Heat Map	Colors the wall with the Bartz wall heat flux
Design Mode	Opens the live sliders (chamber diameter, chamber length — grain length for solids — and expansion ratio)
Orbit / Cam / HQ	Camera behavior, preset viewpoint, and render quality
Reset View	Returns the camera to its starting position



The 3D motor model in cutaway view with dimensions turned on. Every dimension comes from the solver output; the timeline animates how the port grows through the burn.

Assumed dimensions are marked. If the scene had to assume a wall thickness or a grain dimension that the solver did not publish, a badge says so, both on the 3D scene and in the design panel. You do not have to open the 3D view to find out that a drawn dimension was assumed.

The heat-map legend is a design load. The minimum and maximum in the legend are the chamber and throat heat fluxes evaluated at a reference cooled wall. The flux at the equilibrium wall temperature can come out noticeably lower; the two are different questions, and the legend says which one it answers.

5.5 Step 5 — Charts and cross-section

The Performance Analysis Dashboard charts (mass flow rates, pressure distribution, regression rate and port growth, oxidizer injection velocity) are drawn in the application's dark theme and work fully offline. They are computed from the transient port-regression solution, not from a

placeholder curve. Thrust and chamber pressure against time are plotted in their own panel (*Thrust and Chamber Pressure vs Time*) directly below these charts.



The Performance Analysis Dashboard: mass flow rates, pressure distribution, regression rate with port growth, and oxidizer injection velocity. Thrust and chamber pressure against time are plotted in their own panel directly below.

The cross-section is a scaled engineering drawing generated from the same geometry the solver used, with a full legend printed next to it: grey for the chamber wall and forward closure, dark grey for the phenolic liner, brown for the solid fuel grain, tan for the injector plate (the pattern follows the injector type you chose), blue-grey for the nozzle wall, a dashed red circle for the final port diameter at burnout, and cyan dimension lines, all measurements in millimeters.



The motor cross-section with its legend. The injector plate pattern follows the selected injector type, and the dashed red line is the port diameter at burnout.

5.6 Step 6 — Export

Export & CAD Options collects everything you can take out of the program: the 3D model, report PDFs, the Excel workbook, the OpenRocket `.eng` file, STL meshes, and the complete package. Section 10 covers these in detail, including the rules that decide when a part is drawn and when it is left out.

6. Designing a Motor from Scratch

Section 5 drove one motor through the program with the values already chosen. This section is about choosing them. It follows a single decision chain, from a mission statement to a motor the program will publish a verdict on. At every step three things are named: the field you type into, the number the program publishes back, and the band that number has to land in. Where a band is printed here, it is the band the program itself holds; the program prints it in the rejection or the warning, so you never have to take it from this guide alone.

The chain is written for the hybrid page, because that page exposes the most sizing levers. Section 9 says which steps change on the solid and liquid pages.

6.1 Step 0: write the mission down before you open the page

Four numbers define the operating point, and everything else in the form is a consequence of them plus a hardware choice.

Total impulse is the size of the motor in one number. On the *Thrust & Time* tab you enter *Thrust (N)* and *Burn Time (s)* and the page prints the product live as *Calculated Total Impulse*; on the *Total Impulse* tab you enter the impulse directly, between 1000 and 1,000,000 N·s, and may leave thrust or burn time empty for automatic. **Thrust**, accepted from 1 to 10^7 N, fixes the throat area and with it the whole gas path. **Burn time**, accepted 0.1 to 1000 s, fixes the grain

web and the fuel mass. **Design altitude**, entered as *Altitude (m)* between 0 and 50,000, fixes the ambient pressure through the US Standard Atmosphere 1976 and with it the useful expansion ratio; the resulting *Ambient Pressure (bar)* is shown next to it.

Thrust and burn time are not independent of the vehicle. The program sizes a motor for the impulse you ask for; it does not check that the thrust is enough to leave the rail or that the acceleration is survivable. Those are trajectory questions, and the Launch Site page or the trajectory analysis type is where you ask them. Decide the impulse first, then split it into thrust and time.

6.2 Step 1: choose the propellant pair

The pair decides what the program can say about your motor, not just how well it performs. Two selectors, *Fuel Type* and *Oxidizer Type*, drive the thermochemistry, the regression coefficients, and which validation bands apply.

Fuel Type offers HTPB, Polyethylene, PMMA, Paraffin Wax, ABS, PLA, Carbon (Graphite), Aluminum Powder, and two custom entries. *Oxidizer Type* offers Nitrous Oxide, Liquid Oxygen, Hydrogen Peroxide, Compressed Air and a custom entry. The separate oxidizer phase selector (Liquid or Gas) is a state, not a second oxidizer: it decides how the density and the injection model are resolved.

What you gain by staying on a characterized pair. The validation layer holds published bands for a small set of pairs. Where a band exists, an out-of-band result is named with the band printed next to it; where no band exists, the calculation still runs but the check falls back to a generic band and tells you less. The pairs with their own bands are N_2O/HTPB, N_2O/paraffin, and, for specific impulse only, LOX/HTPB.

Pair	Isp band (s)	c^* band (m/s)	O/F band
N_2O / HTPB	180–250	1350–1650	4.0–12.0
N_2O / paraffin	190–260	1400–1700	4.0–12.0
LOX / HTPB	220–280	generic	generic
Any other pair	150–300	1200–1800	0.5–8.0

What the program does when the fuel has no fitted regression law. The regression coefficients a and n in *Fuel Configuration* are filled from the database for the fuel you picked. For a fuel with no validated fit, the warning reads *no validated regression coefficients for this fuel*, states that literature values for a similar fuel were used, and tells you to treat the regression rate, the burn time and the O/F shift as indicative. That warning is the decision point: either accept an indicative answer, or fit your own a and n from static fires and type them into *Fuel Configuration*. The default pair arrives with $a = 3.68 \times 10^{-5}$ and $n = 0.555$, and the fields accept a in 10^{-6} to 10^{-2} and n in 0.1 to 1.0.

6.3 Step 2: choose the mixture ratio

O/F is the first number that has four different answers in the program, and knowing which one you are looking at removes most of the confusion on this page.

Number	Value	What it is
The page default	7.5	The tabulated theoretical optimum for the pair the two selectors start on, N_2O with HTPB. It is a starting value, not a verdict, and it is tagged to that pair
The endpoint band	0.1–50	What the hybrid geometry endpoint will accept before it refuses the request
The canonical band	0.5–20	The repository-wide envelope the endpoint band is derived against
The validation band	4.0–12.0	What raises a warning after the calculation, for this pair

A value inside the endpoint band but outside the validation band calculates and then warns. That is the intended behavior: the first band asks whether the request is physical, the second asks whether the result is inside the range the model was checked against.

Why the optimum sits so high for nitrous oxide. The band's own published basis records that N_2O peaks near O/F 7 to 8 and stays high and flat across roughly 4 to 12, because about two thirds of the impulse comes from the exothermic decomposition of the N_2O itself and only about one third from burning the fuel with the oxygen it releases. Storable oxidizers therefore optimize at high mixture ratio; LOX pairs optimize near 2. If you carry an O/F habit over from a LOX engine you will design a fuel-rich hybrid and see it in both bands at once.

A worked reading. On the default pair, moving O/F from 2.5 to 6.0 moves the specific impulse from 185.4 s to 230.3 s, and an O/F of 2.5 also pushes c^* to 1331 m/s, which is under the 1350 m/s lower edge of the c^* band. At O/F 7.5 both bands are satisfied. If you need a fuel-rich motor for a reason outside the program, such as an ablative liner budget, expect two warnings and read them as the record of a deliberate choice rather than as a fault.

Using *Find Optimum*. Pick both selectors first, then press *Find Optimum*. The pair is sent to the server and the optimum is computed for it. The page prints no "distance from optimum" figure for an arbitrary pair, because it does not know the optimum until it has computed it.

6.4 Step 3: choose the chamber pressure

Chamber pressure buys performance and is paid for twice, once in the tank and once in the wall.

The endpoint accepts 1 to 500 bar on the hybrid page, 5 to 200 on the solid page and 10 to 500 on the liquid page; outside those it refuses the request. The validation check is tighter, 5 to 100 bar, and it raises a warning after the calculation rather than blocking it.

The tank price. The oxidizer has to arrive at the injector above chamber pressure, and the injector has to drop a real fraction of the chamber pressure to stop the feed system from talking to the combustion. The program holds that fraction as a band, and it is the same band the stability panel judges chug with:

$\Delta P_{\text{inj}}/P_c$	What the program says
≥ 0.20	The recommended design rule: the feed system is decoupled from the chamber
0.15 to 0.20	Marginal, the low end of the classical band
< 0.15	Chug risk: feed-to-chamber coupling is possible
> 0.35	Above the classical band. This ceiling is a feed-system cost criterion, not a stability criterion: beyond it you pay tank pressure and pump work for no stability gain

The *Pressure Drop Override (%)* field on the injector tab accepts 10 to 50 and starts at 20. So a 20 bar chamber at the recommended ratio needs about 4 bar across the injector, and the tank

has to stand above 24 bar plus the line and valve losses before any ullage or blowdown margin. The example in Section 5 enters 50 bar, which is a self-pressurizing N_2O storage state rather than a computed minimum.

The wall price. Hoop stress in a thin cylinder is proportional to $P_cD/2t$, so doubling chamber pressure at fixed diameter doubles the thickness the structural module will ask for, and the mass goes with it. This is the trade the *Structural Safety* panel is there to price: run it before and after a pressure change rather than after the design is frozen.

If you leave chamber pressure empty. The hybrid solver still runs, on a declared placeholder, and the placeholder is named in the `defaults_applied` block. It is a placeholder so the arithmetic can proceed, not a recommended pressure. Read the declaration block before you quote any number from such a run.

6.5 Step 4: choose the expansion ratio for the altitude you fly at

Ae/At Ratio accepts `0` for automatic, in which case the solver sizes the nozzle for the ambient pressure that follows from the altitude you entered. When you set it yourself, three checks act on it.

- **The geometry band, 4 to 250.** Outside it the warning prints your ratio and the range. A sea-level design at modest chamber pressure frequently lands under 4, which is why the example motor in Section 5 raises that warning.
- **The separation criterion.** The solver applies a Summerfield criterion: the boundary layer separates when the exit pressure falls below k times the ambient pressure, with k at 0.4 by default and an accepted band of 0.35 to 0.40. At sea level that puts the separation threshold near 0.4 bar of exit pressure. Push the expansion ratio past that and the nozzle is no longer flowing full.
- **What a separated nozzle reports.** Thrust and thrust coefficient are then published on the separated effective expansion ratio, not on the geometric one, and the warning says so. The number is not wrong; it is answering a different question from the one your drawing asks.

The altitude trade in one sentence. A ratio matched to vacuum is unusable at sea level because it separates, and a ratio matched to sea level wastes exit area in vacuum; a first stage is sized for the pressure it spends most of its impulse at, not for the pad. If you want to see the penalty, use the altitude sweep analysis type, which walks the ratio you chose across the altitude interval you set.

A clamped optimum is not a matched nozzle. On the liquid page, if the solved optimum expansion ratio hits the edge of the accepted band, the warning states which edge, prints the clamped value, and says the nozzle is no longer matched to the design altitude. Treat that as a design instruction, not as a rounding note: change the chamber pressure or the design altitude until the optimum falls inside the band.

6.6 Step 5: let the throat follow from thrust

On the hybrid and liquid pages the throat is a result, not an input. Thrust, chamber pressure and the thrust coefficient fix the throat area through $F = C_F P_c A_t$, and the mass flow follows from $\dot{m} = P_c A_t / c^*$. That is why the program refuses a request that carries a quantity it publishes: the throat diameter, the oxidizer mass flow and the fuel mass flow are outputs of that chain, and accepting them as inputs would let you label the solver's own number as your own.

The consequences are worth stating as rules, because they are the fastest way to steer the geometry:

- At fixed chamber pressure, thrust scales with throat area, so with the square of throat diameter. Halving the thrust target shrinks the throat diameter by about 30 percent, not by half.
- At fixed thrust, raising chamber pressure shrinks the throat. That is the mechanism behind "high pressure makes a small engine", and it is also why the wall gets thicker: the same force is carried on a smaller, more highly loaded shell.
- The canonical envelope for a throat is 1 mm to 2 m. A result outside that is a unit error somewhere upstream, almost always a diameter typed in meters into a millimeter field or the reverse.

On the solid page the throat is an input, *Throat Diameter Dt (mm)*, accepted 1 to 500 mm and auto-filled after a calculation. There you set the throat and read the chamber pressure that results from the propellant and the burning area, which is the opposite direction of causation and the reason solid sizing feels different.

6.7 Step 6: size the chamber with L* and the contraction ratio

L Value (m)* is chamber volume divided by throat area. It sets how long the gas stays in the chamber before it reaches the throat, and therefore how completely it burns.

Field	Accepted	Note
L* Value (m), hybrid	from 0.1 m	No upper band is imposed: in a hybrid the useful L* varies very widely with the grain, so an upper limit would refuse legitimate designs. A zero or negative value is refused, because the chamber volume $V_c = L^* A_t$ would then be zero or negative
Characteristic Length L* (m), liquid	0.3–5, starts at 1.2	A tighter band, because a liquid chamber has no grain filling it
Contraction Ratio Ac/At, hybrid	0–10, `0` is automatic	Chamber cross-section over throat area
Contraction Ratio ε_c, liquid	1.5–60, starts at 4	
Chamber Diameter (mm), hybrid	`0` is automatic	Set it and the contraction ratio follows from it

What a low contraction ratio does. The warning names it directly: the chamber Mach number rises, the injector-face pressure climbs above the chamber pressure, and the flow approaches choking inside the chamber. At that point the one-dimensional premise the whole page rests on is straining, and the pressure you typed is no longer the pressure the injector sees. Raise the chamber diameter or lower the contraction ratio target until the warning clears.

Chamber Length Override (mm) and what it may not do. The override replaces the length derived from L*. If your value is shorter than the fuel grain plus the pre-combustion chamber, it is rejected and the warning quotes both your value and the minimum. It is not trimmed to fit, because a trimmed length is a different motor from the one you asked for. Because the chamber cannot be shorter than the grain it contains, the L* the motor actually realizes can be larger than the one you requested; the program reports what it realized rather than echoing your request.

6.8 Step 7: size the grain and the port

This is where a hybrid design is won or lost, and where the program has the most to say.

Three fields drive it. *Initial Port Diameter (mm)* accepts 0 to 500 with `0` meaning automatic: a small port means high oxidizer flux and high regression, and a shorter time to burn through. *Number of Ports* accepts one or more, and empty means one; more ports divide the flow and shorten the grain. *Mass Flux* accepts 0 to 1000 kg/m^2·s, again with `0` for automatic, and is the design oxidizer flux in the port.

The oxidizer flux gate. Port oxidizer mass flux G_{ox} is the oxidizer mass flow divided by the port area. The program warns at 600 kg/m^2·s, which is the lower edge of the band where flooding sets in (600 to 700); it warns early on purpose. Above it, the port cannot take the oxidizer you are pushing through it and the regression law stops describing the surface. The fix is more port area: a larger initial port, or more ports.

The slenderness gate. The program warns when the grain or chamber length-to-diameter ratio exceeds 10. This is a manufacturing and combustion practice band, not a single published constant, and the warning prints the ratio it measured. It matters more than it looks: the default hybrid inputs produce a grain L/D of 18.9 and a chamber L/D of 19.7, which is an 80 mm tube 1.58 m long, and at O/F 2.0 the ratio measures 26.3. Selecting four ports on the same run brings the grain ratio down to 8.6. That is the standard cure, and it is why multi-port grains exist.

What multi-port modeling does and does not cover. The warning is explicit: area and perimeter scaling are correct, but port interaction, web thickness between ports, and burn-through between ports are not modeled. Use the equivalent-port result to size, then check the web between ports by hand.

The web gate. If the port breaks through to the case before the burn time you asked for, the warning says the fuel web is exhausted early and states that performance beyond that point is not physical. Three levers fix it: a larger grain outer diameter, a smaller initial port, or a shorter burn time.

Regression and the feedback you cannot see. The regression law $r = a\,G^n$ is evaluated against the oxidizer flux, and the port grows as it burns, so the flux falls and the O/F shifts through the burn. The charts on the results page plot exactly this: the regression rate with port growth, and the mass flow rates. If the regression iteration does not converge, the warning says the reported rate is the last iterate and not a converged solution; do not quote that run.

6.9 Step 8: choose the injector

Type	What it is for, and what it opens
Showerhead	A simple multi-hole plate, and the default on the hybrid page. Opens the target velocity, the hole diameter limits, the plate thickness and the hole pattern (Circular Rings, Hexagonal, Square Grid)
Impingement	Forces mixing by colliding jets. Opens the pattern (Doublet, Triplet F-O-F, Quadlet, Like-on-Like, Shear Coaxial), the impingement angle, the distance to impingement, the orifice diameter, the element-pair count and the momentum ratio
Pintle	Throttleable and mechanically simple. Opens the pintle tip diameter, the annular gap, the spray angle, the skip distance and the secondary holes
Swirl	Fine atomization at modest pressure drop. Opens the swirl chamber diameter, the target spray half-angle, the tangential slot count, and the slot width and height
Coaxial	For gas-liquid pairs. Opens the inner post diameter, the outer annulus diameter, the recess length and the element count

The accepted range and starting value of every one of these fields is in Appendix C.

Four numbers govern the plate regardless of type, and the injector check holds a band for each: the discharge coefficient C_d (the field accepts 0.20 to 1.0), the hole diameter (0.2 to 5.0 mm),

the injection velocity (10 to 100 m/s), and the orifice Reynolds number (1000 to 200,000). A hole below the small end is not manufacturable and blocks easily; a velocity below the low end atomizes poorly. The plate is then sized structurally from the injector material, and the panel reports the plate bending stress, the safety factor, the required thickness and the mass.

The swirl injector has two design modes and you must pick one. Leave the slot width and height *empty* and the inverse solver designs for the target spray half-angle. Fix the slot geometry and you have fixed the swirl number as well, so the target angle can no longer be honored, and the program says so rather than quietly preferring one of your two instructions. A swirl chamber narrower than 1.25 times the exit orifice breaks the geometric premise of the model and is reported.

6.10 Step 9: choose the wall, the material and the margin

The hybrid page accepts a *Wall Thickness (mm)* of 2 to 20 and starts at 5, with a *Safety Factor* of 2 to 6 starting at 4.0. The solid page accepts a *Case Wall Thickness (mm)* of 2 to 100 and a *Safety Factor FoS* of 1.1 to 10 starting at 2.5; the liquid page accepts a *Chamber Wall Thickness (mm)* of 0.2 to 150 starting at 5 and the same safety-factor band. On all three, the thickness is the hardware and the safety factor is the target.

The distinction that catches everyone. Entering a safety factor does not make the motor safe. Your real wall thickness travels with it, so the module computes the minimum safety factor that pressure, diameter, material and thickness actually deliver, and compares it with your target. The reported minimum safety factor is the answer; the field is the question. A value outside the accepted range is refused with the range named, and a value that cannot be read as a number is refused as such, in both cases without substituting a design value.

The pressure ladder you are designing against. The *Pressure Vessel* panel makes the margin chain explicit, and it is worth designing against the chain rather than against a single number:

Rung	How it is set
MEOP	Your maximum expected operating pressure, filled from the chamber pressure
Proof pressure	1.5 × MEOP, under the AIAA S-080 proof factor
Hydrostatic test	1.3 × MAWP × (S_{test} / S_{design}), under ASME UG-99b
Required burst	2.0 × MEOP under AIAA S-080, or 3.5 × MEOP under the ASME VIII Div 1 interpretation, depending on the *Design Code* selector
Predicted burst	The smaller of the Faupel thick-wall result and the thin-wall plastic limit $P_b = 2\,\mathrm{UTS}\,t/(D+t)$. It is a prediction; no hydrostatic test stands behind it

The weld joint efficiency selector (1.00 full radiography, 0.85 spot radiography, 0.70 none) is a real design lever here, not paperwork: it scales the allowable directly, so an unradiographed seam costs you 30 percent of the wall.

Do not leave the vessel wall thickness blank and then read the green badge. With the thickness blank, the panel sizes the wall to hit its own margin target, and the burst margin it then reports is a read-back of that target rather than an independent verification. The panel says so in the card tip. Type the thickness you intend to build and read the margin you actually have.

The thermal verdict is a separate question, and wall thickness is not its lever. A first honest hybrid design almost always produces two critical thermal messages at once: the chamber wall exceeds the melting point of its material, and the equilibrium wall temperature is pinned within five percent of the adiabatic wall temperature. The second explains the first.

With the cooling selector on *None* the coolant-side film coefficient is natural convection, around 25 W/m^{2}K, which is effectively no path for the heat to leave; a steady-state balance with no outflow has one solution, the wall at the gas temperature, and that exceeds any realistic metal. Thickening the wall does not help, because thickness does not appear in a steady-state balance at all: it changes how fast the wall arrives, not where it stops. The levers that move this verdict, in the order worth trying:

1. **Select a cooling method.** On the hybrid page the *Cooling* selector offers None, Radial Channels and Spiral Channels, and the selection now reaches the thermal model rather than being ignored. This is the only input that changes the outflow term, so it is the only one that can unpin the wall.
2. **Lower the gas temperature.** Moving O/F away from the stoichiometric peak costs specific impulse and buys chamber temperature directly. A few hundred kelvin here is worth more than any amount of steel.
3. **Change the material.** AISI 304 melts at 1673 K, which is below the flame temperature of every useful propellant pair. Inconel 718 buys margin; a graphite or ablative liner changes the problem from conduction to sacrifice.

And read the verdict for what it is. A short-burn amateur motor often survives a wall temperature this model calls fatal, because it never reaches equilibrium: the case absorbs the burn and the firing ends first. The program does not credit that, and says so rather than guessing your burn duration against your wall thickness. Treat the message as a statement about the equilibrium your design is heading toward, and reach for the transient solver, a liner, or thermocouple data before you decide a short burn saves you.

6.11 Step 10: the closing checklist

Run the calculation, then walk this list before you trust the result. Every row is a number the program publishes and a band it publishes with it, so nothing here needs to be looked up.

Check	If it is outside
Warnings panel empty	Work the list top to bottom before you read anything above it
c^* inside the pair band	Thermochemistry or O/F. Check the pair and the mixture ratio first
Isp inside the pair band	If c^* is in band and Isp is not, the nozzle is the suspect: expansion ratio, separation, divergence loss
O/F inside 4.0–12.0 on this pair	Deliberate fuel-rich or oxidizer-rich choices are allowed; record why
Expansion ratio inside 4–250, no separation call	Raise the chamber pressure, or design for altitude and accept the pad penalty
G_{ox} under 600 kg/m^2·s and grain L/D under 10	Larger port, or more ports, which fixes both at once
Web survives the burn	Larger outer diameter, smaller port, or shorter burn
$\Delta P_{\text{inj}}/P_c$ in 0.15–0.35	Below 0.15 is chug risk; above 0.35 you are buying tank pressure for nothing
Minimum safety factor above your target	Thicker wall, stronger material, lower pressure, or smaller diameter
Throat wall under the material limit	Change the nozzle material or add cooling; both temperatures are printed

Change one thing at a time and use *What changed.* Every Analysis Dock panel reports which values moved since the previous run and by how much, and says plainly when nothing changed. That block is the cheapest design tool in the program: it turns a parameter sweep you would otherwise do on paper into two button presses.

7. Reading the Results

A calculation produces numbers, bands, declarations and, sometimes, a refusal to give a verdict. This section is about turning that into a decision. It is organized the way you should read a result: the performance numbers first, because they tell you which half of the motor to look at; then the declarations, because they tell you how much of the answer was computed; then the warnings, because they tell you what the program will not stand behind.

7.1 Three performance numbers, and what each one blames

Specific impulse is the product of the other two: $I_{sp} = c^* C_F / g_0$. That factorization is the single most useful diagnostic on the page, because the two factors describe different hardware.

Quantity	What it measures	What an error in it means
c^*, characteristic velocity	The combustion side only. It depends on the propellants, the mixture ratio and the chamber conditions, and not at all on the nozzle	Thermochemistry, the propellant record, the mixture ratio, or the combustion efficiency you supplied
C_F, thrust coefficient	The nozzle side only: how well the chamber gas is turned into axial momentum	Expansion ratio, contour, divergence, separation, or the ambient pressure you designed against
I_{sp}, specific impulse	The two multiplied, per unit weight of propellant, referenced to the standard g_0	Read it only after you know which factor moved
Total impulse	Thrust integrated over the burn	The size of the motor, not its quality
L^*, characteristic length	Chamber volume over throat area, so residence time	Too short and the gas leaves before it finishes burning, which shows up as a c^* shortfall, not a nozzle problem

The diagnostic in practice. If c^* is inside its band and I_{sp} is low, stop looking at the propellant: the loss is in the nozzle, and the candidates in order of likelihood are a low expansion ratio, a separation call, and divergence loss on a conical contour. If c^* is below its band, the nozzle is irrelevant until you have fixed the chamber: check the propellant pair, the mixture ratio, and whether you supplied a combustion efficiency.

How the program decides between a warning and a critical message. The severity is computed from how far outside the band you are, not from the wording: a specific impulse below 80 percent of the band's lower edge or above 120 percent of its upper edge is critical, and so is a c^* below 90 percent of its lower edge. Inside those factors, an out-of-band value is a warning. This is why a warning and a critical message about the same quantity are not the same statement: the first says the result is outside the checked range, the second says it is outside anything the model can defend.

Combustion efficiency scales everything downstream. If you supply a c^* efficiency on the liquid page (*Combustion Efficiency* η_c, 50 to 100 percent, starts at 97) or on the solid page (0.5 to 1, starts at 0.95), the delivered c^* is that factor times the theoretical value, and the factor then propagates into specific impulse, mass flow and throat area. Measured hybrid efficiencies are typically 0.90 to 0.97. The solver accepts a wider physical band, 0.5 to 1.05, and applies a value outside even that band as given rather than silently correcting it, while recording that it was out of band: an efficiency of 0.300 takes a theoretical c^* of 1328.19 m/s down to a delivered 398.46 m/s, and every number downstream moves with it.

The hybrid page has no such field. The hybrid solver accepts a c^* efficiency only through the Python API and through the Uncertainty analysis, so on the hybrid page the reported c^* is the theoretical equilibrium value and the 0.300 figure above cannot be produced from the interface at all. This matters when you compare a hybrid result with a liquid one: the hybrid

number is an upper bound that no real motor reaches, while the liquid number already carries a 97 percent derate. Subtract roughly three to ten percent from a hybrid c^* before putting it next to a delivered figure from a test stand, and if a hybrid result looks impossibly high, this is the first thing to suspect.

7.2 Reading the flow regime

The *Nozzle Flow* panel classifies the nozzle into one of six regimes, and the label is a design instruction rather than a description:

Regime	What to do about it
`PERFECT EXPANSION`	The exit pressure matches ambient at this altitude. Nothing to fix, but check that it is the altitude you actually fly at
`UNDEREXPANDED`	Exit pressure above ambient: you are leaving thrust in the gas. A larger expansion ratio recovers it, if the vehicle can carry the nozzle
`OVEREXPANDED`	Exit pressure below ambient, but still attached. The loss is real and bounded; this is the normal state of a first stage low in the atmosphere
`SEPARATED`	The boundary layer has left the wall. Thrust is reported on the separated effective ratio. Reduce the expansion ratio or accept that the published number describes a partly filled nozzle
`NORMAL SHOCK IN NOZZLE`	A shock stands inside the divergent section. This is not a flight condition; it usually means the back pressure or the expansion ratio is wrong for the case you meant to run
`UNCHOKED`	The throat is not choked. Almost always a pressure ratio error in the inputs

The same panel prints C_F three ways, and the difference between them is the loss budget: `CF IDEAL` is the loss-free value, `CF` is what the model delivers, and `CF EFFECTIVE` is what survives separation. It also brackets specific impulse between `ISP FROZEN` and `ISP SHIFTING` with a `ISP PREDICTED` value between them, and prints the kinetic loss with the band it was judged against. A predicted value sitting on the frozen end of the bracket means the chemistry has no time to recombine in the nozzle, which is a residence-time statement about your contour.

7.3 What the measured validation spread means for your number

Section 13 publishes the correlation table: the bias and the median absolute percentage error of the model against a database of real experiments. That table is not decoration; it is the error bar you should attach to the number on your screen.

If you are quoting	Measured spread	What to do with the program's number
Hybrid c^*	bias -0.1%, median APE 2.3%, $N = 18$	Use it. A few percent is below the batch-to-batch scatter of real propellant
Hybrid specific impulse	bias $+8.2\%$, median APE 7.7%, $N = 18$	Optimistic by construction: the model predicts theoretical performance and the records carry no nozzle geometry. Subtract the bias before you size a vehicle on it
Hybrid regression rate	bias -24.9%, median APE 36.7%, $N = 35$	Read the split below before you use this at all
Liquid vacuum specific impulse	bias $+0.8\%$, median APE 1.0%, $N = 14$	Excellent agreement, but given the real operating point and published geometry. It is not a from-scratch prediction
Solid burn rate	bias -0.4%, median APE 0.5%, $N = 27$	In-sample: the coefficients are the source author's own fits of those same points. It validates the implementation, not the prediction

The regression-rate row hides two populations, and which one you are in decides everything. Split by source on the published run: the paraffin with gaseous oxygen records ($N = 17$) sit on their own published law with a bias of $+0.4$ percent and a median APE of 4.7

percent, while the low-flux HTPB with N_2O records ($N = 18$) are under-predicted by roughly a factor of two, bias −48.7 percent and median APE 49.3 percent. The cause is documented rather than accidental: a single published pair of coefficients per fuel, fitted in a flux range of roughly 10 to 30 g/cm^2·s, is being evaluated below that range where radiation and small-motor effects dominate. No coefficient was tuned to hide it.

The practical rule. If your port flux is low, expect the program to under-predict regression, which means it will over-predict burn time and under-predict fuel flow, which shifts O/F oxidizer-rich in reality relative to the prediction. If regression rate drives your design, fit a and n from your own static fires and enter them in *Fuel Configuration*. That is the single highest-value input you can give this program.

7.4 The declaration chain: reading a number's history

Every published value carries where it came from. Reading that chain takes three questions, in order.

Question 1: was it computed, or assumed? The value source legend in every Analysis Dock panel tags each input:

`COMPUTED` means it came from the solver run you just did, `USER` that you typed it, `ASSUMED` that the panel applied a default which is then listed by name in the "assumed on your behalf" block, and `NOT MODELLED` that there is no model for it here, so no number was produced.

The ledger below each panel counts the tags, so you can see at a glance how much of a result rests on assumptions. A result built mostly from `ASSUMED` inputs is a screening number.

Question 2: what basis was it judged on? Wherever a number could be mistaken for a measurement or for a different quantity, a basis field travels with it and says which. Four examples you will meet:

- The predicted burst pressure carries a basis stating that it is the smaller of the Faupel thick-wall result and the thin-wall plastic limit. It is a model output. Nothing was burst.
- The specific-heat ratio in the thermal chain carries its source. Leave the field empty and the result says the repository default 1.20 was used, with `gamma_source: repo_default`.
- The injector pressure-drop band carries the origin of each edge: the lower edge is the chug stability minimum, the upper edge is a feed-system cost criterion. They are different kinds of limit and the report says so.
- The proof pressure carries the code factor it came from, so a number computed under AIAA S-080 is not confused with one computed under ASME VIII.

Question 3: was the verdict earned? A number and a verdict are separate products, and the program will publish the first while withholding the second. The structural module makes this explicit: supply your real wall thickness and it runs in verification mode, so the reported minimum safety factor is what your hardware delivers. Leave the thickness out and a sizing routine picks a thickness that meets the target, after which the reported margin reads back that target. Both cases print a number; only one of them is evidence.

> **The test for an earned green.** Ask what would have had to be different for this badge to be red. If the answer is "nothing, the sizer aimed at exactly this", the badge is a read-back. The panels that can produce a read-back say so in the card tip rather than leaving you to work it out.

7.5 The four ways the program declines to answer

They are not interchangeable, and the distinction tells you what to do next.

What you see	What it means	What to do
`NOT MODELLED`	There is no model here for this quantity in this configuration	Nothing in the program will produce it. Compute it elsewhere
`NOT EVALUATED`	There is a model, but it was not run because the load case or the input it needs was not declared	Supply the missing declaration if the form exposes it; otherwise expect this state
"No verdict published"	The number exists and is shown, but the self-consistency gate that would justify a verdict does not cover this run	Read the field data, do not read a pass or a fail into it
"No published data"	There is no citable source for the constant the model would need	Change the material or the propellant, or supply the constant if you have a source

None of these is a defect. Each one is the program declining to produce a number that would not survive being questioned. The list of what this version deliberately withholds is in Section 13, and it is worth reading before you plan an analysis around a verdict that will not arrive.

7.6 Reading the two-phase status on a solid motor

This is the one verdict to check before quoting a solid specific impulse, and it has three distinct states rather than two.

- **The *Two-Phase Loss Factor* field is empty by default** (it accepts 0.8 to 1). Left empty, the solver applies no user two-phase factor and declares that it did not.
- **The published specific impulse therefore carries no verified two-phase correction.** The status appears in the API response, in the band under the specific-impulse cards, in the PDF report, and in the comment header of the `.eng` file, so the qualification travels with the number into whatever flight simulator you hand it to.
- **If the solver publishes no status at all**, the band says *not published by the solver* rather than assuming the correction was applied or was unnecessary.

The module also carries its own linear two-phase loss estimate. It is a diagnostic: reported, never applied. Outside the particle mass fraction window the anchoring reference covers, it is withheld with the reason rather than extrapolated. KN-sugar propellants and black powder fall outside that window; APCP and double base fall inside it. Seeing the diagnostic withheld on a sugar propellant is the model refusing to extrapolate, not a missing feature.

7.7 When two numbers on the same screen disagree

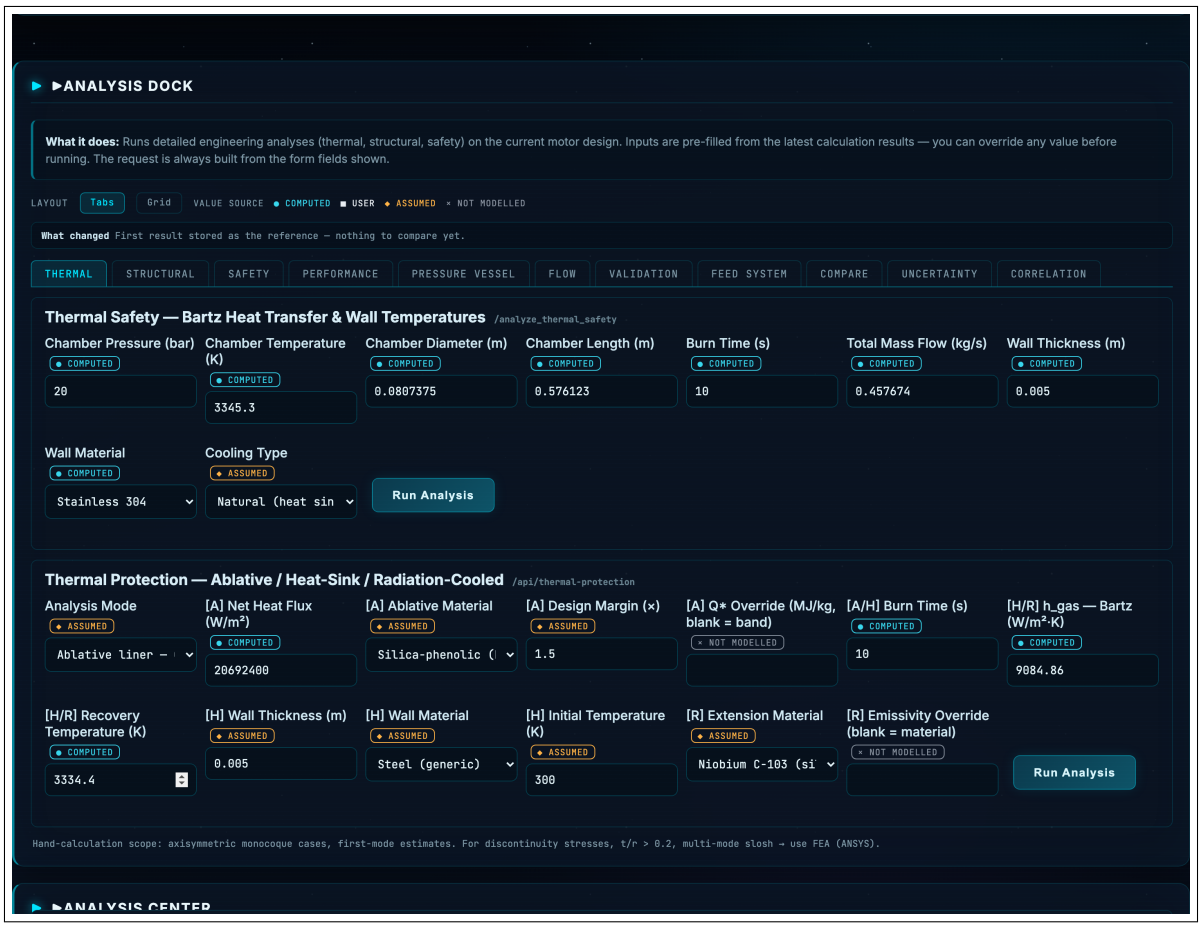
Several quantities appear twice with different meanings, and each pair is explained where it occurs. The heat-map legend on the 3D model is evaluated at a reference cooled wall, which is a design load, while the thermal panel's wall temperature is an equilibrium result (Section 5). The vacuum against sea-level specific impulse gap and the values the liquid solver computes for itself are both covered with the liquid verdicts (Section 9). Web thickness and case free volume are derived on the solid page, so a contradicting entry is quoted back against the geometric value. And the predicted burst pressure is the smaller of two correlations, not a test result.

In every one of these cases the two numbers are answering different questions, and the screen says which. If a pair looks contradictory, the fastest resolution is to read the basis line under the number rather than to recompute.

One reading habit worth building. Read the Warnings panel, then the declaration blocks, then the headline numbers. Read in the other order and you will anchor on a figure before you know how much of it was assumed. Every mechanism in this program is designed so that the qualification of a number reaches you at the same time as the number; the only way to lose it is to skip past it.

8. The Analysis Dock and the Analysis Center

Below the results, two containers hold the deeper analyses. The **Analysis Dock** holds the engineering panels that run on the motor you just calculated. The **Analysis Center** holds the heavier solves, organized as a component-by-analysis matrix.



The Analysis Dock: the category tabs, a panel's pre-filled inputs and the value-source legend. Result plots appear in the panel once you run that analysis.

8.1 How the Dock works

The Dock describes itself on screen: it runs detailed engineering analyses on the current motor design, its inputs are pre-filled from the latest calculation results, you can override any value before running, and the request is always built from the form fields shown. Panels are grouped by category (`FLOW`, `THERMAL`, `STRUCTURAL`, `PRESSURE VESSEL`, `FEED SYSTEM`, `SAFETY`, `PERFORMANCE`, `UNCERTAINTY`, `VALIDATION`, `CORRELATION`, `COMPARE`) and you can lay them out as *Tabs* or as a *Grid*.

Three mechanisms appear in every panel and are worth learning once:

- **The value source legend.** Each input carries a source tag: `COMPUTED` (it came from the solver run), `USER` (you typed it), `ASSUMED` (the panel applied a default, which is listed by

name), or `NOT MODELLED`. The ledger below the panel counts them, so you can see at a glance how much of a result rests on assumptions.

- **The "What changed" block.** After a second run, the panel reports which values moved and by how much relative to the previous run, and says plainly when nothing changed.
- **Refusal to run on blanks.** If a required field is empty, the panel does not send the request. It lists the empty fields and states that a blank field is not a value.

8.2 The panels, one by one

Nozzle Flow (`FLOW`) solves the quasi-1D compressible flow: regime, $P(x)$, $M(x)$ and C_F. Its six required fields are born empty. If you press *Analyze* on a cold panel it still answers, but every default it applied appears by name in the assumed-on-your-behalf block. Arriving from a motor page fills the fields with that motor's real values.

Advanced Performance (`PERFORMANCE`) draws the 3D performance surface, the Mach contour and the heat-flux figure. The relations behind these figures are stated in the last section of the Formulas page.

Structural Safety (`STRUCTURAL`) covers the pressure vessel, buckling and fatigue. Inputs: chamber diameter, wall thickness, material, pressure, design factor. Two things about its verdicts:

- **Buckling is not declared safe unless it was evaluated against a load.** The case buckling block used to compute the axial mode only and report a pass without ever comparing the external-pressure mode with an applied pressure. Unless an external-pressure load case is declared, `buckling_status` now reads `NOT_EVALUATED`, and the API response carries a `buckling_status_scope` field naming the modes that were not covered at all. That field is published in the response body, not printed on the panel, and the load case itself is accepted only by `/api/advanced-analysis`, never by this panel's own endpoint. No page sends it, so on every screen the not-evaluated state is what you get. The panel reference in Section 8 gives the exact keys and what to do about them.
- **"Actual burst pressure" is a prediction, not a measurement.** The number is the smaller of the Faupel thick-wall and the thin-wall membrane limits. No hydrostatic burst test stands behind it. The same value is published as a predicted burst pressure and the basis field says so.

Pressure Vessel (`PRESSURE VESSEL`) sizes the vessel and reports the maximum allowable working pressure alongside the burst prediction described above.

Bolted Joint (`STRUCTURAL`) works the Shigley chain: preload, tightening torque, and the separation margin of the joint.

Thermal Safety (`THERMAL`) runs the Bartz heat-transfer chain and reports wall temperatures. If the thermal path did not run, the wall temperature reads `NOT EVALUATED` instead of showing a stale or assumed number. The ratio of specific heats is checked rather than corrected: a value outside the band the frozen Bartz relations are defined on (1.05–1.67) is refused with the band named. Supply no gamma and the repository default is used, but the result now says that it was a default. The specific-heat ratio (γ) input is accepted in the 1.05–1.67 band; a value outside it is rejected, not clamped. If you leave it empty the repository default 1.20 is used and the result carries `gamma_source: repo_default`.

Regenerative Cooling (`THERMAL`) marches the coolant along the wall station by station (Bartz on the gas side, Dittus-Boelter on the coolant side). If you leave the coolant inlet temperature at

the generic 300 K form default while running a cryogenic fuel, the panel substitutes the cryogenic pump-discharge default and tells you it did.

Thermal Protection (`THERMAL`) covers ablative, heat-sink and radiation-cooled walls. Its ablation chain was rebuilt for this release: the blowing blockage is now solved self-consistently from the mass-transfer number instead of being held at a constant that belongs to a different flow regime, and its derivation is published next to the result. When the solution finds no net heating, the panel does *not* publish a 0.0 mm liner as "sized": zero recession is not zero thickness, because the liner is then governed by a case and bond-line temperature limit that this level of model does not cover, and the reason is stated.

Feed System (`FEED SYSTEM`) covers slosh, pressurant sizing and water hammer. The water-hammer fluid map covers seven propellants, each with primary-source citations for bulk modulus, density and vapor pressure. Hydrogen peroxide is deliberately absent: no citable vapor pressure was obtained, so no water-hammer block is produced for it and no number is invented. Blowdown pressurant sizing applies the real-gas compressibility factor, which matters at high storage pressure. If the water-hammer peak exceeds the pipe hoop yield pressure, the warning names both numbers and tells you to slow the valve closure or upgrade the line.

Comprehensive Safety (`SAFETY`) produces the hazard picture: explosive, toxic and fire hazards, required protective equipment, emergency response and operational safety. Note the panel's own scope statement: the software does not evaluate regulatory compliance, and NFPA, OSHA, DOT and local requirements have to be assessed by a qualified safety authority for your specific site, propellant and operation.

Uncertainty Quantification (`UNCERTAINTY`) runs a Monte Carlo sweep and reports confidence bands rather than a single number.

User Data Validation (`VALIDATION`) compares your own static-fire CSV against the HRMA prediction. Three things to know:

- The relative difference is measured against *your measurement*: the published convention is (prediction minus measurement) divided by measurement, and the response states which quantity is the reference so you read it from the data rather than from memory.
- **Only scored quantities are colored.** The endpoint scores the impulse difference and the normalized RMS error; peak thrust, mean thrust and burn time are not scored and are shown without a color. If the endpoint sends no band, no card is colored: the panel does not invent thresholds.
- An uploaded file is accepted up to 100,000 rows. A longer file is not silently truncated; the upload is refused and the row band is named.

Experimental Correlation (`CORRELATION`) compares the model against the built-in static-fire database, which is the same machinery behind the validation figures in Section 13.

Comparative Analysis (`COMPARE`) snapshots the current configuration so you can put several design points side by side.

8.3 Panel reference: what goes in, what comes out, how to read it

The descriptions above cover each panel's character. This reference covers its mechanics: the inputs it asks for, the numbers it publishes, and the rule for turning those numbers into a decision. Inputs arrive pre-filled from the last calculation wherever the panel can map them; anything you override is tagged `USER` in the ledger.

8.3.1 Nozzle Flow (FLOW)

Reads. Chamber Pressure (bar), Chamber Temperature (K), Gamma (frozen), Molecular Weight (g/mol), Throat Diameter (m), Exit Diameter (m), Ambient Pressure (Pa), Axial Stations (30–60). A *Fidelity Level* selector (Fast Screening, Engineering) and a *Kinetic Model* selector (Fast equilibrium reference, Engineering JANNAF-style); the kinetic branch also reads O/F Ratio, Fuel and Oxidizer.

Publishes. The flow regime label, $P(x)$ and $M(x)$ along the axis, exit Mach, mass flow, c^*, thrust and effective thrust, C_F with its ideal and effective values, specific impulse bracketed between frozen and shifting, and the kinetic loss with its band.

Reading rule. Start with the regime label, because it decides whether the rest is a design point or a diagnostic. Then read the gap between `CF IDEAL` and `CF`: that is your total nozzle loss budget. Then read `CF EFFECTIVE` against `CF`: any gap there is separation.

Refuses. Nothing is sent while a required field is blank; the panel lists the empty fields instead. Its six required fields are born empty, so a cold run applies defaults and names every one of them.

8.3.2 Advanced Performance (PERFORMANCE)

Reads. Chamber Pressure (bar), Optimal O/F Ratio, Base Isp (s), Expansion Ratio, Throat Area (m^2), Nozzle Length (m), Chamber Length (m), Burn Time (s), Base Heat Flux (W/m^2), Critical Heat Flux (MW/m^2), Chamber Temperature (K), Gamma (frozen), Total Mass Flow (kg/s), Chamber Diameter (m).

Publishes. The performance surface in three dimensions, the Mach contour, and the heat-flux figure.

Reading rule. This panel answers "where on the ridge am I" rather than "what is my number". Read the surface for the gradient: a design sitting on a steep face is sensitive to the parameter along that axis, and a design on a plateau is not. The relations behind the three figures are stated in the last section of the Formulas page.

8.3.3 Structural Safety (STRUCTURAL)

Reads. Chamber Pressure (bar), Chamber Diameter (m), Chamber Length (m), Throat Diameter (m), Burn Time (s), Thrust (N), Wall Thickness (m), Material, and three optional thermal inputs: Gas Temperature (K, blank skips the thermal terms), hot-side and cold-side wall temperature.

Publishes. Sections titled *Basis of this verdict*, *Wall temperature behind the thermal terms*, *Wall Stresses*, *Safety Factors*, *Wall Thickness*, *Buckling (NASA SP-8007)*, *Fatigue*, and *Fasteners & End Cap*

Reading rule. Read *Basis of this verdict* before the verdict. It says whether the thermal terms were included at all, which changes the stress state materially: leaving the gas temperature blank gives you a pressure-only answer, which is not conservative for a thin hot wall.

Withholds. `buckling_status` reads `NOT_EVALUATED` unless an external-pressure load case is declared, and the API response carries a `buckling_status_scope` field naming the modes that were not covered at all (local and panel buckling, stiffened-shell and cutout effects, combined bending plus pressure). That sentence is published in the response body, not printed on the panel. The load case is accepted only by the `POST /api/advanced-analysis` endpoint (`analysis_types: ["structural"]`), passed inside `motor_data` as `external_pressure` and

`internal_pressure_min` in pascals. The Structural Safety panel endpoint does not read these two keys, and no page sends them, so on every screen the not-evaluated state is what you get.

What that means in practice. A motor case that is safe against internal pressure can still collapse under external pressure, and a hybrid or liquid stage sees exactly that load during tank drawdown and, on a high flyer, from aerodynamic pressure on an unpressurized section. The program is not telling you the case is safe in that mode; it is telling you it never looked. If your vehicle has an unpressurized section, size that mode yourself against NASA SP-8007 rather than reading the pass on this panel as coverage.

8.3.4 Pressure Vessel (PRESSURE VESSEL)

Reads. Design Code (AIAA S-080 for flight hardware, ASME VIII Div 1 for ground and test), MEOP (bar), Inner Diameter (mm), Wall Thickness (mm, blank auto-sizes), Material, Design Temperature (K), Weld Joint Efficiency (UW-12: 1.00 full radiography, 0.85 spot, 0.70 none), Head Type (2:1 Ellipsoidal, Torispherical, Hemispherical).

Publishes. Predicted burst pressure, burst margin, required thickness, wall used, MAWP, the full pressure ladder, head thicknesses under ASME UG-32 geometry, and the temperature derating.

Reading rule. Read MAWP against your MEOP first: if MAWP is below MEOP the vessel is not qualified for the pressure you plan to run, and the card turns red on exactly that comparison. Then read the thickness margin, which is the ratio of the wall used to the wall required.

Watch for. With the wall thickness blank the vessel is sized to hit the margin it then reports, so the margin is a read-back of the sizing target. The card tip says so. Negative verdicts are still printed in the auto-sized case, because a failure there is real information.

8.3.5 Bolted Joint (STRUCTURAL)

Reads. Chamber Pressure (bar), Seal / Effective Diameter (mm), Bolt Count, Bolt Size, Property Class, Grip Length (mm), Member Material, Thread Condition, Connection Type, and an optional Extra Axial Load (N).

Publishes. Recommended torque, preload F_i, proof safety factor, overload factor n_L, separation factor n_0, plus the load-sharing breakdown and the bolt data.

Reading rule. The separation factor is the one that decides whether the joint leaks: it is the margin before the members come apart under pressure, and a joint can have an acceptable proof factor and still separate. Read n_0 first, then n_L, then the proof factor.

Note. The member material is required; the endpoint will not assume one for you.

8.3.6 Thermal Safety (THERMAL)

Reads. Chamber Pressure (bar), Chamber Temperature (K), Chamber Diameter (m), Chamber Length (m), Burn Time (s), Total Mass Flow (kg/s), Wall Thickness (m), Wall Material, Cooling Type.

Publishes. h_g on the gas side, the throat and chamber heat fluxes at a reference cooled wall, the inner and outer wall temperatures, the adiabatic wall temperature, an axial profile from chamber to nozzle exit, a heat load and cooling assessment, and the thermal safety factors.

Reading rule. The adiabatic wall temperature is the ceiling the wall would reach with no cooling at all, so it bounds the problem. The inner wall temperature is the number to compare against the material limit. The two heat fluxes are design loads evaluated at a reference cooled wall, which is why they can exceed the flux at the equilibrium wall temperature.

Refuses. A specific-heat ratio outside 1.05–1.67 is rejected with the band named, not clamped, because the frozen Bartz relations are not defined outside it. If the thermal path did not run, the wall temperature reads `NOT EVALUATED` instead of a stale number.

8.3.7 Regenerative Cooling (THERMAL)

Reads. Chamber Pressure (bar), Chamber Temperature (K), Throat Diameter (m), Expansion Ratio, Gamma (frozen), Molecular Weight (g/mol), Coolant, Coolant Mass Flow (kg/s), Coolant Inlet T (K), Coolant Inlet P (bar), Number of Channels, Channel Width (mm), Channel Height (mm), Liner Thickness (mm), Liner Material, Flow Direction.

Publishes. Peak wall temperature, wall margin, coolant exit temperature, coolant temperature rise, pressure drop, peak heat flux from the coupled wall balance, maximum coolant velocity, minimum Reynolds number.

Reading rule. Four numbers have to pass together and they trade against each other. Wall margin says the liner survives; coolant temperature rise says the coolant does not boil; pressure drop says the pump can deliver it; minimum Reynolds says the Dittus-Boelter correlation is still valid. Narrowing the channels improves the first and worsens the third.

Watch for. Leaving the coolant inlet temperature at the generic 300 K form default with a cryogenic fuel makes the panel substitute the cryogenic pump-discharge default and tell you it did.

8.3.8 Thermal Protection (THERMAL)

Reads. An *Analysis Mode* selector, then the fields for that mode. Ablative: net heat flux (W/m^2), ablative material, design margin, an optional Q^* override, burn time. Heat sink: h_{gas} from Bartz, recovery temperature, wall thickness, wall material, initial temperature, burn time. Radiation cooled: h_{gas}, recovery temperature, extension material, an optional emissivity override.

Publishes. Ablative: required liner thickness, total recession, mean recession rate, the Q^* used. Heat sink: inner and outer wall temperature at end of burn, service limit, time to limit, margin to limit. Radiation cooled: equilibrium wall temperature, margin, radiated flux.

Reading rule. In heat-sink mode, *time to limit* against your burn time is the whole verdict: a wall that reaches its service limit before burnout has no margin no matter how good the end-of-burn temperature looks. In ablative mode, compare the recession against the thickness you can actually machine.

Withholds. When the solution finds no net heating, no liner thickness is published. Zero recession is not zero thickness: the liner is then governed by a case and bond-line temperature limit that this level of model does not cover, and the reason is stated.

8.3.9 Feed System (FEED SYSTEM)

Three analyses share one panel and are selected by tab: *Slosh*, *Pressurant*, *Water Hammer*.

Reads. Slosh: tank radius, fill height, axial acceleration, propellant density, baffle width and depth ratios. Pressurant: architecture (regulated or blowdown), pressurant gas, propellant volume, gas temperature, and then either tank pressure, storage pressure, regulator margin and collapse factor for the regulated branch, or initial ullage volume, initial pressure and polytropic exponent for the blowdown branch. Water hammer: line fluid, line length, line inner diameter, wall thickness, working pressure and its reference, mass flow, valve closure time, pipe material.

Publishes. Slosh: first-mode frequency, slosh mass fraction and mass, pendulum length, fill ratio, and ring-baffle damping. Pressurant: delivered gas with isothermal and adiabatic bounds, bottle volume and count, residual gas, and for blowdown the ratio, initial and final pressure, final temperature and trapped gas. Water hammer: the surge peak against the pipe hoop yield pressure.

Reading rule. In the pressurant tab the isothermal and adiabatic results bracket reality; size the bottle on the pessimistic bound rather than on the mean. In the water hammer tab, if the peak exceeds the pipe hoop yield pressure the warning names both numbers, and there are exactly two cures: slow the valve closure or upgrade the line.

Withholds. The water-hammer fluid map covers seven propellants with primary-source citations for bulk modulus, density and vapor pressure. Hydrogen peroxide is deliberately absent because no citable vapor pressure was obtained, so no water-hammer block is produced for it.

8.3.10 Comprehensive Safety (SAFETY)

Publishes. The pressure-vessel summary as a pressure ladder (operating, design, proof test, hydrostatic test, burst target), individual risks scored 1 to 5, a 5 by 5 risk matrix, a mitigation priority list, a *Regulatory compliance* section, and a block listing the inputs the endpoint filled from its own defaults.

Reading rule. Read the defaults block first: on this panel more than any other, a risk score computed from assumed inputs is a category statement rather than a site assessment. The mitigation priority list is the actionable output.

Scope. The panel states its own limit: the software does not evaluate regulatory compliance. NFPA, OSHA, DOT and local requirements have to be assessed by a qualified safety authority for your specific site, propellant and operation.

8.3.11 Uncertainty Quantification (UNCERTAINTY)

Publishes. A run summary carrying the confidence level, the number of samples, the number of failed samples, the random seed, the elapsed time, the sampler and the core used, followed by the output distributions and the sensitivity ranking.

Reading rule. Read the failed-sample count before the bands. Samples fail when the solver could not complete for that draw, and a high failure count means the bands were computed on a censored population. Then read the sensitivity ranking rather than the bands: it tells you which input to measure better, which is the only action a confidence band supports.

The nominal is not the centre of the band, and that is deliberate. The panel pins each nominal to the value the deterministic *Calculate* run used, so the two screens agree. Several input distributions are centered somewhere else: the hybrid c^* efficiency, for instance, is sampled from a truncated normal with mean 0.93 on $[0.80, 1.00]$, while the nominal is pinned at 1.0 because the main hybrid route applies no efficiency at all. The gap is published rather than hidden, as `mean_shift_percent` per output. Read it: a -7 percent shift on specific impulse is not noise and not an error, it is the panel telling you that the deterministic number on the main page is the optimistic end of the distribution, and that the honest expectation is seven percent below the headline. Where you see a large negative shift, quote the shifted figure in a design review, not the headline.

Note. The seed is published so the run is reproducible. Quote it when you quote a band.

8.3.12 User Data Validation (VALIDATION)

Reads. Your static-fire CSV.

Publishes. The thrust curve with your test data against the HRMA prediction, a parsed-data summary, and the agreement cards: total impulse, impulse difference, peak thrust, mean thrust, burn time, RMSE and NRMSE.

Reading rule. Only the impulse difference and the normalized RMS error are scored, so only those two are colored. Peak thrust, mean thrust and burn time are shown uncolored because there is no threshold behind them. The relative difference is measured against your measurement: prediction minus measurement, divided by measurement, and the response states which quantity is the reference.

Refuses. A file longer than 100,000 rows is refused with the row band named rather than silently truncated.

8.3.13 Experimental Correlation (CORRELATION) and Comparative Analysis (COMPARE)

Experimental Correlation runs the model against the built-in static-fire database and prints the main layer as a motor-type by quantity matrix, the same machinery behind the validation figures in Section 13 and subject to the same three cautions. *Comparative Analysis* snapshots the current configuration and highlights the best thrust, specific impulse and efficiency across the snapshots: it is the tool for the trade study the design chapter keeps asking for, so change one parameter, calculate, snapshot, repeat.

8.3.14 The nozzle CFD run card

The Analysis Center rows and their verdicts are covered in the four subsections below. Only the CFD run card needs its inputs listed, because it is the one card whose settings change what the solve can be asked to prove.

It reads the chamber stagnation pressure and temperature, the ratio of specific heats, the specific gas constant, the ambient pressure, an optional back pressure, an optional Summerfield k, the grid resolution (Coarse or Standard), the wall model (inviscid, viscous isothermal, viscous adiabatic), the turbulence model (none for laminar, or Spalart-Allmaras in its negative variant), and, for an isothermal wall, the wall temperature. Its result sections cover the inlet state, the measured against expected comparison, the convergence criteria, grid quality, transport, the boundary flux budget, the viscous cost, friction, and the declarations. Read them in that order: a converged solve on a bad grid is not a result.

8.4 The Analysis Center

The Analysis Center asks two questions: *where* (which component) and *what* (which analysis). Pick a row on the left, fill the run card in the middle, and read the result on the right. Rows that do not apply stay visible in grey with the reason named, rather than disappearing.

Component	Analysis	Motor types
Chamber + nozzle wall	Structural (linear static, axisymmetric)	hybrid, solid, liquid
Chamber + nozzle wall	Thermal (transient conduction, Bartz BC)	hybrid
Solid grain cross-section	Plane strain (SP-8073 acceptance)	solid
Nozzle internal flow	CFD (Euler, shock capturing, separation call)	hybrid, solid, liquid
Tank / bolted joint / line	Module cards (pressure vessel / bolted joint / water hammer)	hybrid, solid, liquid
Chamber acoustics	Acoustic mode table	hybrid, solid, liquid
Chamber acoustics	Combustion stability (feed-coupled chug loop)	hybrid, solid, liquid

Each row carries a state: `READY`, `AWAITING RESULT` (run the motor calculation first), `NOT APPLICABLE` (with the reason, for example that the analysis does not apply to this motor type), or `NOT IN CENTER` (the analysis has not been moved here yet and still runs in its own panel). Runs are kept in a per-session history strip; a result recalled from history is labelled as stored rather than fresh.

8.5 Reading the CFD verdict

The nozzle CFD run prints a row of badges. They are worth reading in order, because they tell you whether the verdict above them was earned:

- **Convergence** is a three-valued verdict. `CONVERGED` with an iteration count, `NOT CONVERGED` with the residual, or **"no verdict published"** when the run falls outside the scope of the self-consistency gate. A residual plateau is called out separately: if the scheme residual stood still for many iterations, a bigger iteration budget does not earn the verdict, and the badge says so.
- **Separation** is judged only when the criterion applies. You may see a separation prediction with the fraction of the divergent length affected, no separation with the margin, "separation criterion not applicable", or an explicit statement that the bridge refused the solution or that the judgment is suspect because the field it was applied to did not settle.
- **Inviscid or viscous.** An Euler run states that it has no wall shear and no wall heat flux. A viscous run names the turbulence model and the wall treatment, and its energy budget badge says whether the wall heat and viscous boundary fluxes are inside the budget.
- **Wall resolution.** The first-cell y^+ badge tells you whether the wall is resolved, in the buffer layer, or in the log layer; in the last case the badge states that the wall heat flux reads low.

Only a firm verdict is shown in green. A run outside the scope of the self-consistency check therefore never receives a positive verdict; the field data are still published, only the verdict is withheld.

8.6 Acoustics and combustion stability

The *Acoustic mode table* reports nozzle damping and the total loss budget per mode. This path publishes no stability verdict, only damping numbers against thresholds, and it says so with its own badge. The numbers are shown in a neutral color precisely because there is no threshold behind them to earn a color.

The *Combustion stability* (chug) panel needs the feed-line length to include the inertance term. Measured example (25 kN LOX/RP-1, 60 bar, O/F 2.2): without a feed-line length the loop reports "unstable" at 77.4 Hz; with a 2.5 m line it reports "stable" at 33.6 Hz. One input changes the verdict, so enter the length when you know it.

The *Combustion stability* row runs the feed-coupled chug loop, and one input changes the physics rather than the precision:

The feed-line length changes the model, not just the answer. Without a feed-line length the loop stays in the form without inertance and declares that in its basis field. Supply a length and the inertial term enters. On one measured liquid stage the same engine reads *unstable* with no line length and *stable* with a 2.5 m line. HRMA does not assume a plumbing layout on your behalf; if the length matters to you, enter it.

8.7 The finite-element panels

Three FEA panels sit outside the Dock, anchored to the pages that can use them:

- **Axisymmetric wall structural FEA** (hybrid and liquid): von Mises stress field, safety-factor field, element-quality map and convergence history. If one of the four plots is missing, the reason is reported: a missing material record blocks the safety-factor field, and a solve that converged in a single round has no history to plot.
- **Transient wall thermal FEA** (hybrid): the wall temperature field, the inner-wall history and the time history. Its ambient temperature field arrives pre-filled. When the bridge cannot model your case, the refusal is now the answer: the request is rejected and the missing input is named, instead of a refusal arriving inside a "success" envelope. A reader that treated a successful HTTP status as "the analysis ran" has to read the verdict itself.
- **Plane-strain grain FEA** (solid): the acceptance criterion is the port bore strain, not the peak von Mises stress, and the badges say so.

Two ways a finite-element verdict is withheld. If the closure base placement margin exceeds the mesher's own thickness tolerance, the safety factor is still published but *no verdict is issued*, and the reason is listed. If the thin-wall premise breaks (the thickness-to-radius ratio exceeds the thin-wall limit), the panel shows a thin-wall badge. When the ratio could not be measured at all, no warning is printed rather than a reassuring one.

9. Solid and Liquid Motors

The solid and liquid pages follow the pattern of Section 5. What changes is the input set and a handful of verdicts specific to each motor type.

9.1 Solid motors

The solid page is organized in nine numbered sections: *0. Propellant Catalogue*, *1. Propellant Properties*, *2. Grain Geometry*, *3. Combustion Chamber and Casing*, *4. Nozzle*, *5. Efficiency and Empirical Coefficients*, *6. Initial & Environmental Conditions*, *7. Mass & Weight Breakdown*, and *8. Measurement / Test Equipment*.

The propellant catalogue is the fastest way to start. Filter by family (composite, sugar KN-based, double base, other), pick a row, and its validated properties load into the form. Three rules govern it, and they are printed on the page:

- Every loaded value stays editable; the catalogue only fills the fields.

- A field you type into is marked as overridden and is never overwritten by a later catalogue load. The page counts your overrides and offers either *Reset to catalogue values* or *Keep my edits*.
- The burn rate law is stated in the units the solver uses: $r\,[\mathrm{m/s}] = a \times (P\,[\mathrm{bar}])^n$.

An untouched field no longer overrides the catalogue. The form used to send a burn-rate coefficient even when you had not typed one, which defeated the catalogue resolution on the server and produced a different thrust curve together with a spurious off-catalogue warning. Untouched fields are not sent; the server resolves the catalogue value and declares where it came from.

Grain geometry offers BATES (cylindrical), star, finocyl, slotted, wagon wheel and end burner, with the number of grains, outer diameter, core diameter and grain length. Web thickness is a derived output, not an input: if you enter a web thickness that disagrees with the value implied by the geometry, the solver uses the geometric value and the warning quotes both. The same rule applies to the case free volume.

The two-phase correction status is the one verdict to read before quoting a solid Isp. Under the Isp cards the page prints a status band, and it is honest about three different situations:

- The two-phase loss factor field is *empty by default*. Left empty, the solver applies no user two-phase factor and declares that it did not.
- The published Isp therefore carries **no verified two-phase (particle lag) correction**. The status appears in the API response, in the band on the page, and in the PDF report, and the warning tells you plainly not to read that Isp as a verified delivered performance.
- If the solver publishes no status at all, the band says *not published by the solver* rather than assuming that the correction was applied or was unnecessary.

A diagnostic that is now withheld outside its anchor window. The module's own linear two-phase loss estimate is a diagnostic: it is reported, never applied. For propellants whose particle mass fraction falls outside the window the anchoring reference figure covers, the number is now withheld with the reason named, instead of being extrapolated. KN-sugar propellants and black powder fall outside that window; APCP and double base fall inside it and are unchanged.

Two further changes affect the numbers you will read. The thermochemistry of the KNDX and KNSB records now comes from the same two-phase equilibrium basis as the other sugar propellants, so the four sugar records share one verdict class; the measured effect is a small rise in c^* and in delivered Isp. And a propellant with no record in HRMA is not silently approximated: the warning names the propellant, lists the known ones, and states that the grain mechanics and two-phase results are not specific to your propellant.

9.1.1 Choosing the grain geometry

The geometry decides the shape of the thrust curve, and the program will not choose it for you. Six entries, and the choice is made on burning-area history rather than on manufacturing convenience:

Grain Type	Burning-area history	Where it fits
BATES (Cylindrical)	Burns outward from the core and inward from the two ends. With several segments the ends compensate the growing core	The default starting point. Simple to cast and to model
Star	The star points burn away first, so the area can be held nearly constant	When you want flat pressure without segmenting
Finocyl	A cylindrical bore with fins at one end	High initial area with a usable web; a common compromise
Slotted	Slots cut through the web	Raises the initial area sharply
Wagon Wheel	Many radial spokes	Very high initial area, short web
End Burner	Burns as a disc along the axis	Long burn at low thrust; the smallest area of all

The geometry fields are *Number of Grains N*, *Outer Diameter (mm)*, *Core Diameter (mm)* and *Grain Length (mm)*, plus the shape-specific fields: star point count, point radius and fillet radius; fin count, width and length; slot count, width and depth. Face inhibition is selected separately for the front face, the rear face and the outer surface, and it changes the burning area directly, so it changes the pressure curve as much as the shape does.

Web thickness and case free volume are outputs. Both fields exist, and both are derived from the geometry. If what you type disagrees with what the geometry implies, the solver uses the geometric value and the warning quotes both numbers. Use the fields to check your understanding of the grain, not to override it.

9.1.2 A worked solid design

This is the page's own starting configuration, stated as a design so you can see which number drives which. Load a catalogue row first; the values below are what the form arrives with.

Field	Value	Why it is the driver
Propellant Name/Formulation	APCP	Sets density, flame temperature, molecular weight and the burn-rate law
Density, flame temperature, molecular weight, γ	1850, 3200, 28.5, 1.25	Density with the grain volume fixes the propellant mass; flame temperature with molecular weight fixes c^*
Number of Grains N	3	Segmenting flattens the pressure curve
Outer Diameter / Core Diameter (mm)	100 / 30	The difference is the web, so it sets the burn time
Grain Length (mm)	500	With the core, it sets the burning area
Chamber Pressure (bar)	40	Accepted 5–200 on this page
Throat Diameter D_t (mm)	15	An input here, auto-filled after a calculation
Expansion Ratio ε	5.44	Accepted 1–25
Case Material / Yield Strength	Steel / 250 MPa	With the case thickness, it sets the margin
Initial Temperature (K)	298	Feeds the temperature coefficient σ

What to expect, and in which order to read it. The burn rate follows $r\,[\mathrm{m/s}] = a \times (P\,[\mathrm{bar}])^n$, so chamber pressure and burning area are coupled through the throat: a larger burning area raises the pressure, which raises the burn rate, which raises the area again on a progressive grain. Read in this order.

1. **The pressure curve.** If it climbs steeply, the grain is progressive; either segment it, inhibit a face, or change the shape.
2. **The two-phase status band under the specific-impulse cards.** Until you read it you do not know whether the specific impulse is a delivered number or a theoretical one.

3. **The temperature coefficient.** The initial temperature is a real design input on a solid motor, not an environmental note: σ is accepted up to 0.02 per kelvin, and a cold-soaked motor and a sun-soaked one are different motors.
4. **The case margin.** Case thickness is accepted from 2 to 100 mm, the yield strength from 10 to 3000 MPa, and the safety factor from 1.1 to 10. The structural verdict uses your thickness, not your target.

Small solid motors are over-predicted. Below roughly 75 mm the delivered specific impulse can be substantially optimistic, because two-phase flow, heat loss and short-L^* scale effects are not modeled. The 100 mm grain above sits just above that line; a 54 mm research motor does not.

9.2 Liquid engines

The liquid page is organized in nine numbered sections: *1. Propellant Data (Fuel + Oxidizer)*, *2. O/F Optimization Data*, *3. Feed System / Engine Cycle*, *4. Injector System Parameters*, *5. Combustion Chamber Design*, *6. Nozzle & Cooling System*, *7. Startup & Transient Analysis*, *8. Mass & Structural Parameters*, and *9. Environment & Test Conditions*.

The engine cycle is the choice that reshapes the rest of the page: *Pressure-Fed*, *Gas Generator*, *Staged Combustion*, *Full-Flow Staged Combustion*, *Expander Cycle* and *Tap-Off Cycle*. A power-balance solver checks whether the pump and turbine powers close, and the thermochemistry is solved at the engine's real chamber pressure, mixture ratio and expansion ratio.

Read these verdicts carefully on this page:

- **The vacuum Isp band.** The interface checks how far the vacuum Isp sits above the sea-level value and compares it against the band expected from the ambient-pressure term of the thrust coefficient. Outside that band the warning prints both Isp values, the percentage, the accepted band and the relation it comes from, and tells you to check the expansion ratio and the ambient pressure you entered before using those two numbers.
- **A capped expansion ratio is not a matched nozzle.** If the solved optimum expansion ratio is clamped at the edge of the accepted band, the warning states which edge, what the clamped value is, and that the nozzle is no longer matched to the design altitude.
- **Feed pressure against the chamber.** A pressure-fed cycle whose tank pressure is below what the lines and injector need is named as such, with both the entered and the required pressure. The same check runs against the required pump discharge pressure in a pumped cycle.
- **Where the tank pressure is read.** In a turbopump-fed cycle the injector is fed by the pump, not by the tank: the tank is pressurized only for pump suction margin, and the solver sizes it itself and publishes it. A tank pressure sent into that cycle would change no published number, so the request is refused and the field you should be using is named.
- **Pump suction margin.** If the required pump NPSH exceeds what is available at the assumed tank pressure and suction-line losses, the warning gives both values and the two ways out (raise tank pressure or add an inducer). Where no vapor-pressure record exists for the propellant, the chain says which saturation assumption it used instead of a measured value.
- **Dump cooling publishes nothing it does not model.** Selecting dump cooling used to hand you the regenerative jacket's flow and pressure drop under a regenerative label. There is no bleed-fraction model for dump cooling, so the flow and pressure drop are now empty, the channel count is zero, and the provenance states that the bleed is not modelled.

- **A non-converged power balance has no residual.** The residual is the solver's own convergence remainder, not an independent cross-check, and when the cycle does not converge the field is empty with the reason given.
- **Staged-combustion pump head.** The required pump discharge pressure is sized on the main stage. If you compare against an older release for the same design, expect a lower discharge pressure; the power split was always right, the head was not.

Entered values that are comparison only. Some liquid fields are computed by the solver from the injector pressure drop, discharge coefficient and mass flow. If you enter one of them, the warning tells you which value the solver computed and states that your entry is reported for comparison only. It is not silently ignored, and it does not silently win.

9.2.1 Pressure-fed or turbopump: the choice that reshapes the page

The *Engine Cycle Type* selector is the first decision on the liquid page, and it changes which fields are read at all.

Cycle	What feeds the injector	What the page then reads
Pressure-Fed	The tank	Tank pressure has to exceed chamber pressure plus the injector drop plus the line losses. The check names both the entered and the required pressure
Gas Generator	A pump driven by a turbine on a separate gas stream	Generator Gas Temperature (K), Turbine Inlet Pressure (bar), Turbine Expansion Ratio, Turbopump Efficiency (%)
Staged Combustion	A pump driven by a preburner whose flow re-enters the chamber	The same turbine fields, plus a *Preburner Type*. Pump discharge pressure is sized on the main stage
Full-Flow Staged Combustion	Two preburners, both flows re-entering	As above
Expander Cycle	A pump driven by coolant heated in the jacket	The cooling fields become part of the power balance
Tap-Off Cycle	A pump driven by gas tapped from the chamber	As for the gas generator

In a turbopump cycle the tank pressure is not an input. The injector is fed by the pump, and the tank is pressurized only for pump suction margin, which the solver sizes itself and publishes. A tank pressure sent into that cycle would change no published number, so the request is refused and the field you should be using is named. This is the single most common liquid-page rejection.

The power balance is what decides whether the cycle closes. If it does not converge, the residual field is empty with the reason given, because the residual is the solver's own convergence remainder and not an independent cross-check. There is nothing to read into an empty residual except that the cycle did not close.

9.2.2 A worked liquid design

The page's starting configuration, reduced to the entries that drive the result. Appendix C carries the rest of the form.

Field	Value	Note
Fuel Type / Oxidizer Type	RP-1 / LOX	The other fuels are LH2, Methane, MMH and UDMH; the other oxidizer is N2O4
Stoichiometric O/F	3.4	The reference point, not the design point
O/F Range Minimum / Maximum	2.0 / 3.2	The interval the optimizer searches
Target Mixture Ratio (O/F)	blank	Leave it empty to let the optimizer choose inside that interval
Chamber Pressure P_c (bar)	blank	Accepted 10–500
Target Thrust (N)	blank	Accepted 100 to 10^7
Characteristic Length L^* (m)	1.2	Accepted 0.3–5
Contraction Ratio ε_c	4	Accepted 1.5–60
Nozzle Expansion Ratio ε_e	50	A vacuum-class ratio; at sea level expect a separation call
Injector Pressure Drop	20	**In bar, not percent**, accepted 0.2–200 bar. At P_c = 20 bar that is a ratio of 1.0, far above the classical band
Number of Injector Elements	100	With the orifice diameters, it sets the injection velocities
Cooling Type	Regenerative	The alternatives are Ablative Liner, Film Cooling, Radiation Cooling and Dump Cooling
Coolant Inlet Temperature (K)	300	A generic default: change it for a cryogenic coolant
Maximum Wall Temperature (K)	800	The limit the cooling solution is judged against
Chamber Wall Thickness (mm)	5	Passed to the structural verdict as built, not as target
Structural Safety Factor	2.5	The target the verdict is compared with
Feed System Pressure (bar)	130	Read differently in each cycle

The injector pressure drop is entered in bar on this page and as a percentage on the hybrid page. The solver reads the liquid field as bar, so entering 25 at a chamber pressure of 20 bar drives the required oxidizer tank pressure to 53 bar. The unit is printed on the field and the rejection states the band the solver applies. Convert deliberately: at P_c = 20 bar the recommended ratio of 0.20 is 4 bar, not 20.

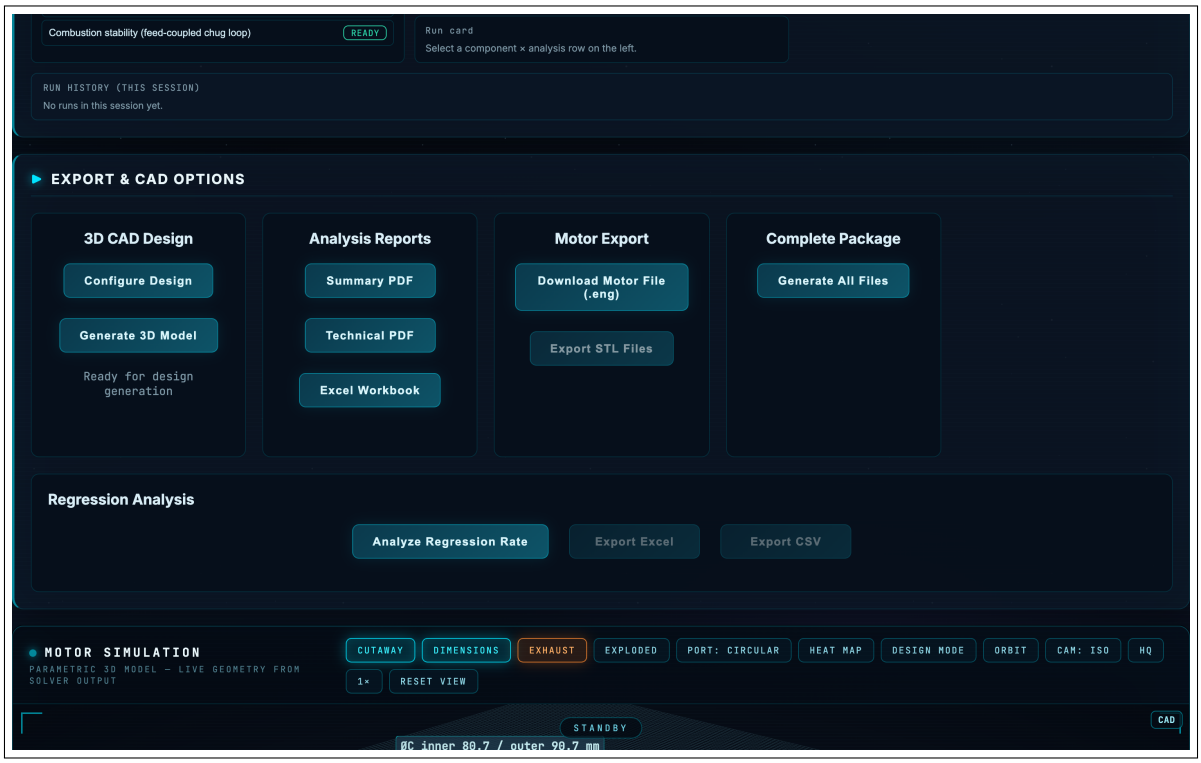
Reading the cooling verdict. The cooling type changes what the solver will publish, and one option publishes deliberately less than the others. Regenerative cooling gives you channel flow, velocity, pressure drop and a wall margin. Dump cooling gives you none of those: there is no bleed-fraction model for it, so the flow and pressure drop are empty, the channel count is zero, and the provenance states that the bleed is not modeled. That is not a gap in the page, it is the model declining to publish a regenerative jacket's numbers under a dump-cooling label.

Reading the pump verdicts. Three of them travel together:

- **Required pump discharge pressure** against your feed pressure. In a staged-combustion cycle it is sized on the main stage; if you compare against an older release for the same design, expect a lower discharge pressure than before.
- **NPSH available against NPSH required.** If the requirement exceeds what the tank pressure and the suction-line losses provide, the warning gives both values and the two ways out: raise the tank pressure, or add an inducer. Where no vapor-pressure record exists for the propellant, the chain states which saturation assumption it used instead of a measured value.
- **The vacuum against sea-level specific impulse gap.** The page checks the gap against the band implied by the ambient-pressure term of the thrust coefficient and, outside it, prints both values, the percentage, the band and the relation, and tells you to check the expansion ratio and the ambient pressure before using either number.

10. Exports and Reports

Export & CAD Options appears under the results once a calculation exists. It has five groups.



The Export & CAD Options panel: 3D CAD Design, Analysis Reports, Motor Export, Complete Package and Regression Analysis.

Group	Buttons	What you get
3D CAD Design	Configure Design, Generate 3D Model	The design configuration panel, and the interactive model built from solver geometry
Analysis Reports	Summary PDF, Technical PDF, Excel Workbook	The report documents and the data tables
Motor Export	Download Motor File (.eng), Export STL Files	An OpenRocket-compatible motor file, and the mesh files
Complete Package	Generate All Files	One archive containing the CAD, drawings, report and motor file
Regression Analysis	Analyze Regression Rate, Export Excel, Export CSV	The regression-rate study and its data

Beyond these, the results panel carries *Export Results*, and the CAD chain can produce a STEP package, a DXF profile and a technical-drawings PDF. All output files land under `Documents/HRMA`, with CAD files in the `cad_exports` subfolder. The export buttons stay disabled until there is a result to export. If a calculation then fails, the results panel is hidden again and the export buttons are switched back off: the previous run's numbers are not left on screen next to your new inputs, and you cannot export a file that describes a motor you no longer have.

10.1 What the exports refuse to draw

The same fail-closed rule governs the exports, and it is the reason an export sometimes contains fewer parts than you expected:

- **The injector plate is drawn only with an earned thickness.** HRMA does not derive the plate thickness on its own; it comes from the injector panel. If it did not, the plate is left out of both the STL and the STEP package rather than drawn at a substituted thickness. This is why a complete liquid-engine package can now be produced where it previously failed: the correct behavior is to omit the part, not to refuse the whole package or to invent the dimension.
- **Geometry that the mesh does not represent is named.** An export rejection lists the fields it is missing, with a reason and the value you submitted for each, and where several addresses would be acceptable it lists them. A machine-readable code travels with the rejection and the English sentence travels next to it.
- **Downloads are direct.** In this release every export file is downloaded straight to your computer; there is no server-side link that expires later. If a file does not download, the message names the reason (missing geometry, an unearned dimension, a rejected field) and asks you to generate the package again. The application never invents a lifetime or a deadline.

10.2 What the reports contain

The PDF report covers the design, the performance, and the safety assessment. That last section is worth a note: it has carried a heading for a long time, but nothing filled it in, so it read as not evaluated and all six hazard blocks were dropped. The server now runs the safety analysis itself, once, from exactly the motor data the report renders, so the document and the verdict cannot come from two different runs. Explosive, toxic and fire hazards, required protective equipment, emergency response and operational safety all reach the document, and a provenance line states which field each input was resolved from.

The same rule closes the other way: if the report carries no usable chamber pressure, propellant mass, chamber diameter or wall thickness, *no verdict is produced* and the missing measure is named. A safety verdict is not computed from assumed dimensions.

The `.eng` file carries the two-phase status. The delivered Isp in the motor file is accompanied by the two-phase correction status in the file's comment header, for solid, hybrid and liquid motors alike. If you hand the file to a flight simulator, the qualification of the Isp travels with it.

If the PDF fails its own check. The report is verified after it is built. If that check fails, HRMA downloads a plain-text report instead and tells you it did, rather than handing you a PDF it could not verify.

11. Input Rules and Error Messages

This section explains what the program refuses, why, and what the refusal looks like on screen. Most support questions come down to one of these rules.

11.1 An empty field is not a zero

Empty fields are not sent to the server. The point is that the server can then distinguish "the user left this out" from "the user set this deliberately", and answer accordingly: apply a canonical default and name it, or refuse and name the field it needs. Three concrete consequences:

- The eight vehicle and environment fields on the hybrid page (vehicle mass, reference area, drag coefficient, launch angle and altitude, wind speed and direction, oxidizer density) go unsent when empty rather than as an explicit null.
- **A parametric sweep with no interval is refused.** If the start or end of the sweep is missing, no points are computed and the missing field is named. A curve drawn over an interval the server guessed but labelled with your parameter name would describe nothing. The step count, by contrast, has a canonical default, and the endpoint declares it.
- The oxidizer density field is filled by the server. If it cannot be, the field stays empty with a hint naming the source, rather than carrying a static number that matches no storage state.

11.2 Values are checked against published bands

Out-of-band values are rejected with the band named; they are not clamped into range. The bands come from one table, so the form and the endpoint cannot drift apart:

Quantity	Accepted band	Where it applies
Chamber pressure (hybrid)	1–500 bar	Hybrid preliminary geometry and the optimum O/F search
Chamber pressure (solid)	5–200 bar	Solid motor calculation and the plane-strain grain solve
Chamber pressure (liquid)	10–500 bar	Liquid engine calculation
Thrust (hybrid)	1–10^7 N	Hybrid preliminary geometry
Thrust (liquid)	100–10^7 N	Liquid engine calculation
Burn time	0.1–1000 s	Hybrid preliminary geometry
O/F ratio	0.1–50	Hybrid preliminary geometry
Mixture ratio (liquid)	0.5–20	Liquid engine calculation
Chamber temperature	300–6000 K	Hybrid geometry and the advanced heat-transfer branch
Ambient temperature	180–340 K	Quick geometry and the thermal FEA
Ratio of specific heats	1.05–1.67	Bartz heat-transfer chain

A field that genuinely has no defensible band says so rather than carrying an invented one, and a rejection body states the band it applied together with the basis of that band.

11.3 A computed result is not an input

Some names appear both as an input elsewhere and as an output here. Sending the oxidizer mass flow into the main hybrid calculation is refused, because that endpoint *publishes* the oxidizer mass flow: it is a result, and accepting it would let you label the solver's own number as your input. The rejection says where the quantity is published and where to go if you wanted to set it.

11.4 Number-like text is not a number

The program accepts JSON numbers. It does not accept a number wrapped in quotes, and it does not accept values that are not finite. All the main calculation endpoints use the same vocabulary for the same defect, so the message you get does not depend on which page you were on:

What you sent	Reason reported
`NaN` or infinity	the value is not finite
A number in quotes, such as `"20.0"`	the value is not a JSON number
`true` or `false`	a boolean is not a number
Text that is not a number at all	the field name with the reason

The tank temperature is a good example of why this matters. A single request carrying a temperature as text once gave one value to the motor and a different one to the solver, a split of more than forty kelvin, with no warning at all. The temperature is now resolved once, together with its alias, and text is refused.

11.5 What a rejection looks like on screen

When the server refuses a request, the page shows the server's own words, assembled in a fixed order:

```
[machine_code] The sentence explaining the refusal · field (reason), field
                        (reason) · missing fields
```

Nothing in that line is written by the page. If the body cannot be read at all, you get the bare HTTP status and nothing more, because a plausible but invented reason would be worse than none. When a panel refuses to run at all, it names the empty fields and states that the request was not sent.

11.6 Warnings you are likely to meet

Message	What it means
c^* outside the expected band	Your characteristic velocity is outside the range expected for this propellant combination; the band is printed
Expansion ratio outside the expected band	The area ratio is outside the range the check covers
Flow separation in the nozzle	The exit pressure is far below ambient; thrust is reported on the separated effective ratio, not the geometric one
Contraction ratio is low	The chamber Mach number rises, injector-face pressure climbs above chamber pressure, and the flow approaches choking in the chamber
Regression iteration did not converge	The reported rate is the last iterate, not a converged solution
No validated regression coefficients for this fuel	Literature values for a similar fuel were used; treat regression rate, burn time and O/F shift as indicative
The fuel web is exhausted early	The port breaks through to the case before the requested burn time; performance beyond that point is not physical
Multi-port grain modelled as equivalent ports	Area and perimeter scaling is correct, but port interaction, web thickness and burn-through are not modelled
Throat over the allowable temperature	The throat wall exceeds the material's allowable temperature with the selected cooling; both numbers are printed
Injector plan not reported	The circuit model could not size the selected injector; do not take injector geometry from this run
Input out of range, catalog value used	Your entry did not reach the solver; the built-in value was used and the accepted range is printed

12. Common Mistakes and How to Recognize Them

Section 11 states the rules. This section is the field guide: what you actually see on screen, what it means, and the fix. Every entry is a case the program really refuses or really warns about.

12.1 Refusals: the request never reached the solver

What you see	What happened	The fix
A rejection naming *ambient temperature* with the band 180–340 K	The ambient temperature is the motor hardware's own envelope, not the chamber temperature. It does not share the chamber band	Enter an atmospheric value. If you meant the combustion temperature, that is a different field with a 300–6000 K band
A rejection saying the value is a computed output of this endpoint	You sent a quantity the endpoint publishes, such as the oxidizer or fuel mass flow	Remove it. The rejection names where the quantity is published and, where a settable equivalent exists, names that field
A rejection naming the field and offering a different one to use	You used a name the endpoint does not read, such as an alias for chamber pressure that this endpoint ignores. The field was being accepted and then silently dropped	Use the name the rejection gives you
A rejection on the tangential entry angle	The swirl solver has a solvable envelope; the field accepts 45 to 71.2 degrees. An angle above it used to be clipped to the edge without telling you	Enter an angle inside the envelope, or design by fixing the slot geometry instead
not a JSON number, *not finite*, or *a boolean is not a number*	A number arrived in quotes, or as `NaN`, an infinity, or a boolean	Type a bare number. A temperature that once arrived as text gave one value to the motor and another to the solver, a split of more than forty kelvin, with no warning
tank pressure is not read in a turbopump cycle	You entered a tank pressure on a pumped liquid cycle	Remove it. The tank is pressurized for pump suction margin only, and the solver sizes and publishes that itself

Every refusal is assembled from the server's own words, in a fixed order:

```
[machine_code] The sentence explaining the refusal · field (reason), field (reason) · missing fields
```

Nothing in that line is written by the page. If the body cannot be read at all you get the bare HTTP status and nothing more, because an invented reason would be worse than none.

12.2 Accepted, but not what you meant

These produce a result. The result describes a different motor from the one you had in mind, and something on screen says so.

Symptom	Cause	The fix
The injector pressure drop is an order of magnitude off	The liquid page reads that field in **bar**; the hybrid page reads its own override in **percent**	Convert deliberately. At 20 bar chamber pressure the recommended ratio of 0.20 is 4 bar, not 20
A dimension is out by a factor of 1000	Millimeters entered in a meter field or the reverse. Chamber diameter appears in millimeters on the form pages and in meters in several Dock panels	Read the unit in the field label. The canonical envelopes are wide enough that a factor of 1000 can still land inside them
A green badge on a vessel you have not sized	The wall thickness was left blank, so the vessel was sized to hit the margin it then reported	Enter the thickness you intend to build
The chug loop says unstable and you expected stable	The feed-line length was not supplied, so the loop ran without the inertance term	Enter the length. On one measured liquid stage the same engine reads unstable with no line length and stable with a 2.5 m line
A cryogenic coolant with a room-temperature inlet	The coolant inlet temperature was left at the generic 300 K form default	The panel substitutes the cryogenic pump-discharge default and tells you it did. Enter your real inlet temperature so the substitution is not needed
No injector geometry in the results	The circuit model could not size the type you selected, so the hole count, hole diameter and manifold diameter are not reported rather than estimated	Read the warning: do not take injector geometry from that run. Change the injector type or supply the missing parameters

12.3 Warnings that are design instructions

Section 11 lists what each warning means. The rows below add what it is asking you to decide, for the warnings where the decision is not obvious from the wording.

Warning	The decision it is asking for
c^* outside the expected band	Fix the chamber before you touch the nozzle: propellant pair, mixture ratio, combustion efficiency. A nozzle change cannot repair a c^* shortfall
O/F outside the expected band	Either move toward the optimum for this pair, or record why you are running fuel-rich or oxidizer-rich on purpose
Expansion ratio outside 4–250	Raise the chamber pressure, or accept a sea-level-matched nozzle and its lower specific impulse
Flow separation in the nozzle	Reduce the expansion ratio, or design for altitude knowingly and read the effective ratio rather than the geometric one
Oxidizer flux near the flooding limit	The warning fires at 600 $\mathrm{kg/m^2{\cdot}s}$, the lower edge of the 600–700 band. Larger port or more ports
Grain or chamber slenderness above 10	Add ports. Measured on the page defaults, four ports take the grain ratio from 18.9 to 8.6
The fuel web is exhausted early	Larger outer diameter, smaller initial port, or shorter burn. Performance past the breakthrough is not physical
A capped expansion ratio	The optimum was clamped at a band edge, so the nozzle is no longer matched to the design altitude. Change the pressure or the altitude
Regression iteration did not converge	Do not quote that run. The reported rate is the last iterate

12.4 The habit that prevents most of the above

Read the unit in the label every time, not once. The same physical quantity appears in millimeters on a form page and in meters in a Dock panel, and the injector pressure drop is a percentage on one page and a pressure on another. The label is always right; your memory of the other page is not. After every change, read *What changed* before the headline number: a change that moved

nothing usually means the field you edited was not read by that endpoint.

13. Scope and Limits (Please Read)

HRMA is a capable preliminary-design and educational tool. Knowing where it stops is part of using it.

13.1 What the models are

The solvers are closed-form relations and one-dimensional correlations. Turbopump maps, cavitation, rotordynamics, bearing life, startup and shutdown transients, and two- or three-dimensional conjugate heat transfer are not present. The nozzle CFD is an Euler solver with shock capturing; it is not a validated viscous design tool, and it states that in its own badges.

13.2 What has been measured

HRMA carries a correlation runner that scores the model against a database of real experiments and regenerates its own figures. The table below is the main layer of that run, as published in `docs/VALIDATION_STATUS.md`. "Bias" is the signed mean of (prediction minus measurement) divided by measurement; "Median APE" is the median absolute percentage error; N is the number of scored records.

Motor	Quantity	N	Bias %	Median APE %
hybrid	characteristic velocity c^*	18	−0.1	2.3
hybrid	specific impulse	18	+8.2	7.7
hybrid	chamber pressure	35	+16.7	13.5
hybrid	final port diameter	18	−9.4	10.1
hybrid	regression rate	35	−24.9	36.7
liquid	sea-level specific impulse	4	+2.0	2.0
liquid	vacuum specific impulse	14	+0.8	1.0
liquid	vacuum thrust	4	−0.4	0.2
solid	burn rate	27	−0.4	0.5

Read that table with three cautions, all of which the source document states itself:

- **Some rows are not independent.** The solid burn-rate row is scored against the fit's own source data, which makes it an implementation check rather than an independent validation. The vacuum-thrust row carries a curator caveat on every record.
- **Good agreement at a known point is not prediction from scratch.** Reproducing vacuum Isp to about one percent *given* the real operating point and published geometry does not mean HRMA predicts the flight performance of an unknown motor to that error.
- **Hybrid regression rate is the weak link, and the aggregate cell hides which half you are in.** The −24.9 percent figure is the mean of two populations that behave nothing alike. Scored against the Karabeyoglu paraffin/GOX campaign ($N = 17$) the model sits on its own published law: bias +0.4 percent, median APE 4.7 percent. Scored against the Rezaei low-flux HTPB/N_2O campaign ($N = 18$) it under-predicts by roughly a factor of two: bias −48.7 percent, median APE 49.3 percent. That second half is a documented model limit rather than a data error — one published a and n per fuel, a correlation whose validity band is about 10 to 30 g/cm^2·s, and radiation plus small-motor effects that dominate below it. No coefficient was tuned to hide the gap, which is why the aggregate reads worse than a tuned model would.

The design rule that follows: before you trust a regression rate, read your own oxidizer port mass flux off the results. If it is inside the correlation band, the number is trustworthy to a few percent. If you are below it — which is where most student and amateur N_2O/HTPB motors actually sit — assume the program is under-predicting regression. Follow that through the chain: fuel flow $\dot{m}_f = \rho_f A_b \dot{r}$ is under-predicted with it, so the reported O/F is too high and the real motor runs richer than the report says; the port opens faster than predicted, so the real final port diameter is larger; and the web can burn through before the burn time you entered. The independently measured final-port-diameter row in the same table (-9.4 percent) is the same error seen from the geometry side, which is a useful consistency check on this reading. Fit your own a and n from static fires and enter them in *Fuel Configuration*; that is the one input that moves this error, and it is why the field exists.

Outliers are flagged, never dropped, and records the model's own declared scope does not cover are reported in a separate layer: measured and withheld, not unmeasured.

13.3 Known limitations

- **Liquid delivered Isp is idealized.** The liquid figures are optimum-expansion values; real engines deliver noticeably less because of finite expansion ratio and nozzle losses. The thermochemistry is sound; the delivered-Isp figures are optimistic upper bounds.
- **Small solid motors are over-predicted.** Below roughly 75 mm, delivered Isp can be substantially optimistic, because two-phase flow, heat loss and short-L* scale effects are not modelled.
- **Heat transfer is a Bartz-level estimate.** The gas-side coefficient and wall temperature are the right order of magnitude and conservative, but a real burn-through margin needs thermocouple data.
- **Combustion instability, ignition transients and hard starts** are outside the program entirely. Delivered c^* efficiency can only be established by static fire.
- **Two-dimensional CFD with back pressure.** In a nozzle CFD run with a back pressure (`POST /api/cfd/nozzle`, `Pb_Pa`) the outlet boundary condition imposes the back pressure also on cells where the flow reverses. In shocked, over-expanded cases the final field can show backflow at the outlet. These cases already return "not converged / plateau" and a `suspect` separation verdict; none is presented as verified. In these runs treat the wall pressure near the outlet and the separation output as qualitative.
- **Supercritical hydrogen coolant velocity.** Channel sizing also applies the 40 m/s liquid coolant velocity target (NASA SP-8087) to supercritical hydrogen; a hydrogen-specific target is planned for 3.1. The result errs on the safe side: deeper channel, lower coolant-side heat transfer, hotter predicted wall. So it does not produce a false SAFE.
- **Hybrid TNT equivalent in the API and the PDF report.** The derived factor is per kilogram of oxidizer; but since the endpoint and the report carry no oxidizer mass input, the factor is applied to the total propellant mass. The result is high on the cautious side. The safety panel does not evaluate the TNT equivalent for hybrids.
- **Star and finocyl grains are not exported to CAD.** The CAD core cannot yet draw these sections faithfully; the export is refused with the reason.
- **The assembly STL is not a closed solid.** Use STEP for CAD work.

13.4 Not in this release

The following are *not* in v3.0; they are planned for 3.1 and later:

- The C++ solver core; 3D CFD and 3D FEA; reacting (combustion) CFD; modal FEA.
- CAD in hand-drawn detail: injector head and dome, flanges and bolt circle, O-ring grooves, ports, regenerative cooling channels, the real port section, a closed STL.
- Model and CAD of system parts: turbopump, preburner, feed lines, valves, gimbal; exhaust plume visuals. The turbopump power balance in the cycle panel is a cycle calculation, not a pump design.

13.5 Verdicts this version deliberately does not publish

Knowing which verdicts are withheld is as useful as knowing which are given. In this version the following are stated rather than answered:

- **CFD convergence outside the self-consistency gate:** no verdict published.
- **Buckling without a declared external-pressure load case:** not evaluated, with the uncovered modes named.
- **Two-phase correction on solid Isp:** not verified, on every surface that publishes the Isp.
- **The linear two-phase diagnostic outside its anchor window:** withheld with the reason.
- **Throat erosion for a material with no published coefficient band** (tungsten): no published data.
- **Dump cooling jacket flow and pressure drop:** not modelled.
- **Water hammer for hydrogen peroxide:** no citable vapor pressure, so no block is produced.
- **Ablation liner thickness when there is no net heating:** not published, because the governing limit is outside this level of model.
- **A finite-element verdict when the closure base margin exceeds the mesher's tolerance:** withheld while the number is still shown.
- **Regulatory compliance:** not evaluated, in any form.

The golden rule. HRMA is a starting point. Cross-check every critical result with an independent code (CEA / RPA / openMotor) and, where possible, with a static fire. Knowing how to drive the program is not, by itself, being a rocket engine engineer.

14. Automatic Updates and Troubleshooting

14.1 How updating works

At startup, HRMA asks GitHub whether a newer version exists. If one does, a window appears offering **Update now**, **Later**, or **Skip this version**.

Update now downloads the installer with a progress indicator. When the download finishes, HRMA closes and the installation runs; the application reopens by itself when it is done. On macOS this usually takes a few minutes, and the message says so: do not open HRMA by hand while it is working.

The flow is written to survive a bad network:

- If the download is slow or interrupted, the window offers a browser link so you can fetch the installer yourself.
- If the installer cannot be started, the message tells you to open the downloaded file from your Downloads folder.
- If GitHub rate-limits your network, which happens on shared connections, HRMA says so and falls back to the ordinary releases page, which it can still read.
- The window states where the version information came from: the Releases API, or the releases page as a fallback.

Skip this version suppresses that one version. Settings lists the skipped version and *Reset skipped version* makes it available again. *Check for Updates…* in the window menu and *Check now* in Settings both run the check on demand.

14.2 Troubleshooting

Symptom	What to do
The window opens but stays on the loading page	The calculation engines are still loading. If it does not clear, close the window and start HRMA again; the log file under `Documents/HRMA` records what happened
Nothing happens when you start HRMA	A copy is probably already running. HRMA does not open a second instance; look for the existing window
"Could not reach the HRMA server"	The local engine stopped or the request was interrupted. Restart the application
A page shows `HTTP` and a number with no explanation	The server refused the request and the body could not be read. Check the values you entered and retry; the log file has the detail
An export button is greyed out	Either there is no result yet, or you changed the form after calculating. Run the calculation again
An export file did not download	Read the message: it names the missing geometry or the rejected field; fix it and generate the package again
The database badges stay grey	The CEA or NIST check has not answered yet. A grey badge is an unknown state, not a failure
Windows warns about the installer	The package is not code-signed. Use "More info", then "Run anyway"
macOS refuses to open the app	Right-click the application and choose "Open" once
Something is genuinely wrong	Use *Report a bug* in Settings. It builds the report, shows you the exact text, and opens a pre-filled GitHub issue that you submit yourself

A. Appendix A: Menus and On-Screen Controls

A.1 Window menu

Item	macOS	Windows
Settings…	Application menu	*HRMA* menu
Check for Updates…	Application menu	*HRMA* menu
Open Output Folder	Application menu	*HRMA* menu
User Guide…	*Help* menu	*Help* menu
Release Notes…	*Help* menu	*Help* menu
HRMA on GitHub	*Help* menu	*Help* menu
Releases and Downloads	*Help* menu	*Help* menu

A.2 Top bar, on every page

Language, Release Notes, User Guide, Settings. *Report a bug* is not a bar control of its own: it is a section inside the Settings panel.

A.3 Home page only

The version badge, the three *Launch Designer* cards (one per engine type), the Formula Reference link and the Launch Site (3D Earth) link.

A.4 Navigation bar, on the designer and formula pages

Home, Hybrid, Solid, Liquid, Formulas, and the language selector. The entry for the page you are on is shown as the active one.

A.5 Project bar, on the three designer pages

Project name, unsaved-changes indicator, Save, Save As, Open, New, Import from CAD (STEP). Overwrite, Discard changes and Cancel are not bar buttons: they appear in the confirmation dialogs those actions open.

A.6 Designer page action buttons

The three designer pages do not carry the same set. The table below is the one place in this guide where they are put side by side, because a control you remember from one page is often simply absent on another.

Control	Hybrid	Solid	Liquid
Run the solver	Calculate	Calculate Motor Performance	Calculate Engine Performance
Find Optimum (O/F)	yes	—	—
Reset the form	—	Reset	Reset
Reset overrides	—	Reset to catalogue values	—
Configure Design / Apply Design Configuration	yes	—	—
Preview Profile (altitude)	yes	—	—
Export Results (one button)	yes	—	—
Export set (separate buttons)	Summary PDF, Technical PDF, Excel Workbook, Download Motor File (.eng), Export STL Files, Export Excel / Export CSV for the regression data	Excel Workbook, DXF Drawing, STEP Package, Complete Package (ZIP), Motor CAD Export, Download OpenRocket .eng	OpenRocket .eng, Technical Drawings PDF, DXF Drawing, STEP Package, Complete Package (ZIP), Excel Workbook
Analysis tabs	Single Point Analysis, Parametric Analysis, Trajectory Analysis	—	Single Point Analysis, Parametric Analysis, Trajectory Analysis
Run the trajectory	Calculate Trajectory Performance	Compute Trajectory	inside the Trajectory Analysis tab
Run the parametric sweep	Run Parametric Analysis	—	inside the Parametric Analysis tab
Ask AI to Analyze Results	yes	yes	yes

Two asymmetries are worth remembering because they are easy to read as faults. *Find Optimum* exists only on the hybrid page, since it is the oxidizer-to-fuel optimum for a propellant pair and the solid page has no such pair to optimize. *Export Results* is a single hybrid button that packages the run; the solid and liquid pages expose the same material as separate format buttons instead, so there is nothing missing there, only a different shape.

A.7 3D viewer

Cutaway, Dimensions, Exhaust, Exploded, port shape, Heat Map, Design Mode, Orbit, camera preset, render quality, playback speed, Reset View. Design Mode adds live sliders for chamber diameter, chamber length (grain length for solids), and expansion ratio. The solid and liquid pages build their viewer toolbar from a shared deck component, which adds a Frame button and, when CFD output is present, a CFD FIELD group with its own Close button.

A.8 Analysis Dock

Layout: Tabs or Grid. Category tabs: FLOW, PERFORMANCE, THERMAL, STRUCTURAL, PRESSURE VESSEL, FEED SYSTEM, SAFETY, UNCERTAINTY, VALIDATION, CORRELATION, COMPARE. Per panel: Run, the value-source legend, the value ledger, and the "What changed" block.

A.9 Analysis Center

Component tree on the left, run card in the middle, result viewer on the right, run history along the bottom. *Reload from current result* refills the card from the latest calculation.

B. Appendix B: Glossary

Term	Meaning as used in HRMA
Thrust	The force the motor produces, in newtons.
Total impulse	Thrust integrated over the burn, in newton-seconds; the size of the motor in one number.
Specific impulse (Isp)	Impulse per unit weight of propellant, in seconds, referenced to the standard g_0 (not the local gravity the Launch Site page reports).
Characteristic velocity (c^*)	The combustion side of performance, independent of the nozzle. Errors here are thermochemistry errors.
Thrust coefficient (C_F)	The nozzle side of performance. Isp is the product of the two divided by g_0.
Chamber pressure	The absolute pressure in the combustion chamber, in bar throughout the interface.
O/F ratio	Oxidizer-to-fuel mass ratio. Called mixture ratio on the liquid page.
Expansion ratio (ε, Ae/At)	Nozzle exit area divided by throat area.
Contraction ratio (Ac/At)	Chamber cross-section divided by throat area.
Characteristic length (L*)	Chamber volume divided by throat area; sets how long the gas stays in the chamber.
Nozzle	The convergent-divergent duct: throat, divergent section, exit. Contour options are conical, bell and parabolic.
Divergence loss	The thrust lost because the exit flow is not purely axial.
Regression rate	How fast a hybrid fuel surface recedes, usually written $r = a\,G^n$ against the oxidizer mass flux G.
Burn rate	The solid-propellant equivalent, written $r = a\,P^n$ against chamber pressure in bar.
Web thickness	The propellant thickness that has to burn through; on the solid page it is derived from the grain geometry, not entered.
Port	The cavity the gas flows through inside the grain; its diameter grows during the burn.
Grain	The cast solid propellant or hybrid fuel body.
Two-phase loss	The performance lost because condensed particles lag the gas. In HRMA it is a diagnostic unless you supply a factor, and its status is always published.
Bartz correlation	The standard relation for the gas-side heat transfer coefficient; the basis of the thermal results.
NPSH	Net positive suction head, the margin that keeps a pump from cavitating.
Chug	The low-frequency feed-coupled instability the combustion stability panel evaluates.
Water hammer	The pressure surge produced when a valve closes on a flowing line.
Fail closed	The program's habit of refusing, withholding or declaring rather than substituting a value it cannot justify.

C. Appendix C: Field Reference

Field names below are written exactly as they appear on screen in the English interface. The second column gives the range the field itself enforces, the value it starts at, and what the program does when you leave it empty. Where the endpoint holds a different band for the same quantity it is the endpoint band that a rejection quotes; the canonical envelopes behind those bands are at the end of this appendix. An empty field is never transmitted.

C.1 Hybrid Motors page

Field	Accepted, starting value, and what a blank does
Motor Name	any characters; used in export filenames. Blank uses the default name
Altitude (m) above sea level	0–50,000, starts at 0
Ambient Pressure (bar)	0–1.2, starts at 1.013; computed from the altitude
Thrust (N)	from 1, starts at 1000
Burn Time (s)	from 0.1, starts at 10
Total Impulse (N·s)	1000–1,000,000, starts at 10,000; on the *Total Impulse* tab only
Chamber Pressure (bar) absolute	from 1, starts at 20; a declared placeholder is applied and named
Tank Pressure (bar) absolute	from 1; empty
O/F Ratio	from 0.1, starts at 7.5; the default belongs to the N_2O/HTPB pair
Fuel Type	10 entries, starts at HTPB
Oxidizer Type	5 entries, starts at N2O
Temperature (K), oxidizer	starts at 293; sets the oxidizer density
Density (kg/m^3), oxidizer	from 100; empty; filled by the server; if it cannot be, the field stays empty with a hint naming the source
Viscosity (Pa·s), oxidizer	starts at 0.0002

Field	Accepted, starting value, and what a blank does
Regression Rate Coefficient (a)	10^{-6}–10^{-2}, starts at 0.0000368; database value for the fuel
Regression Rate Exponent (n)	0.1–1.0, starts at 0.555; database value for the fuel
L* Value (m)	from 0.1; empty; chamber length derived from the default
Initial Port Diameter (mm)	0–500, 0 is auto, starts at 0
Number of Ports	from 1; empty; one port
Ae/At Ratio, 0 is auto	from 0; empty; sized for the design altitude
Chamber Diameter (mm), 0 is auto	from 0; empty; derived
Contraction Ratio Ac/At	0–10; empty; derived
Mass Flux (kg/m^2·s)	0–1000, 0 is auto, starts at 0
Oxidizer Feed Line Length (m)	0.05–50; empty; the chug loop runs without the inertance term and declares it
Oxidizer Feed Line Inner Diameter (mm)	0.5–500; empty
Nozzle Type	Conical, Bell, Parabolic, starts at Conical
Combustion Type	Infinite Area, Finite Area, starts at Infinite
Injector Type	Showerhead, Impingement, Pintle, Swirl, Coaxial, starts at Showerhead

Field	Accepted, starting value, and what a blank does
Injector Discharge Coefficient	0.20–1.0; empty
Pressure Drop Override (%)	10–50, starts at 20
Number of Holes Override	1–200; empty; computed
Plate Thickness (mm)	starts at 3.0
Chamber Material	AISI 304, Aluminum 6061-T6, Inconel 718, starts at AISI 304
Wall Thickness (mm)	2–20, starts at 5
Safety Factor	2–6, starts at 4.0
Chamber Length Override (mm)	empty; derived from L*
Nozzle Material	Graphite, Tungsten, Copper (Regeneratively Cooled), starts at Graphite
Injector Material	AISI 316, Titanium Grade 5, Brass (Low Pressure), starts at AISI 316
Include Cooling Channels	None, Radial, Spiral, starts at None
Ambient Temperature (K)	180–340; empty

Injector-specific fields. These appear only for the type you select.

Type	Field	Accepted / starts at
Impingement	Impingement Pattern	Doublet, Triplet (F-O-F), Quadlet, Like-on-Like, Shear Coaxial
	Impingement Angle (degrees)	30–120, starts at 60
	Distance to Impingement (mm)	2–30, starts at 10
	Orifice Diameter (mm)	0.3–3, starts at 0.8
	Number of Element Pairs	4–50, starts at 8
	Momentum Ratio (Fuel/Ox)	0.5–2.0, starts at 1.0
Pintle	Annular Gap (mm)	0.5–5, starts at 2
	Spray Angle (degrees)	20–90, starts at 45
	Skip Distance (mm)	5–50, starts at 15
	Secondary Holes	None, Radial, Tangential
Swirl	Swirl Chamber Diameter (mm)	5–50, starts at 20
	Target Spray Half-Angle (degrees)	45–71.2, starts at 71.2
	Number of Tangential Slots	2–12
	Slot Width (mm) / Slot Height (mm)	1.0 / 2.0. Leave both empty to design for the target angle
Coaxial	Inner Post Diameter (mm)	2–20, starts at 5
	Outer Annulus Diameter (mm)	3–30, starts at 8
	Recess Length (mm)	−5 to +10, starts at 0
	Number of Elements	1–100, starts at 19

C.2 Solid Motors page

Field	Accepted range and starting value
Propellant Name/Formulation	catalogue or typed, starts at APCP
Density ρ_p (kg/m^3)	500–3000, starts at 1850
Heat of Combustion ΔH_c (kJ/kg)	500–10,000, starts at 2500
Adiabatic Flame Temperature T_c (K)	1000–4500, starts at 3200
Molecular Weight M (g/mol)	2–100, starts at 28.5
Specific Heat Ratio γ	1.05–1.5, starts at 1.25
Characteristic Velocity c^* (m/s)	800–2500, starts at 1550
Burn Rate Coefficient (a), m/s/bar^n	starts at empty; the catalogue value is resolved on the server
Burn Rate Exponent (n)	−0.5–1.0; blank by default
Erosive Burning Coefficient k	0–1, starts at 0.0002
Erosive Burning Exponent m	0.1–2, starts at 0.8
Temperature Coefficient σ (1/K)	0–0.02, starts at 0.002
Grain Type	six entries, starts at BATES
Number of Grains N	from 1, starts at 3
Outer Diameter (mm)	1–5000, starts at 100
Core Diameter (mm)	5 to (chamber diameter − 5), starts at 30; refused with HTTP 400 outside it for radially burning grains, unbounded for an end burner

Two different refusals hide in that table. *Core Diameter* is a hard gate: outside its band the request is refused outright and the answer quotes the band it computed for your chamber. The four material rows above it — density, adiabatic flame temperature, molecular weight and outer diameter — behave the opposite way: outside the band the field is *ignored*, the catalogue value for the propellant is used instead, and the answer says so. The run still succeeds, which is exactly why it is worth knowing: a density of 4000 kg/m^3 does not stop anything, it quietly becomes the catalogue density and every mass in the report belongs to a different propellant than the one you typed.

Field	Accepted range and starting value
Grain Length (mm)	starts at 500
Web Thickness (mm)	derived, starts at 35
Grain-to-Grain Gap (mm)	starts at 2
Insulation Thickness (mm)	0–100, starts at 3
Chamber Diameter (mm)	10–5000 (endpoint), starts at 100
Case Internal Volume (L)	0.01–16,000, starts at 2.5
Case Material	Steel, Aluminum, Composite, Titanium, starts at Steel
Yield Strength σ_y (MPa)	10–3000, starts at 250
Safety Factor FoS	1.1–10, starts at 2.5
Case Wall Thickness (mm)	2–100; blank by default
Liner Thickness (mm) / Density	0.1–100 / 200–3000, starts at 2 / 1200
Initial Temperature (K)	200–350, starts at 298
Throat Diameter D_t (mm)	1–500, starts at 15
Exit Diameter D_e (mm)	1.5–2500, starts at 35
Expansion Ratio ε	1–25, starts at 5.44
Convergent / Divergent Angle (°)	10–80 / 5–45, starts at 45 / 15

Field	Accepted range and starting value
Nozzle Efficiency η_n	0.8–1, starts at 0.95
Erosion Factor (mm/s)	0–5, starts at 0.001
Nozzle Material	Graphite, Phenolic, Tungsten, starts at Graphite
Chamber Pressure (bar)	5–200 (endpoint), starts at 40
Combustion Efficiency η_c	0.5–1, starts at 0.95
Nozzle Discharge Efficiency (Cf)	0.8–1, starts at 0.98
Overall Thrust Efficiency	0.5–1, starts at 0.92
Kinetic Efficiency	0.8–1, starts at 0.97
Divergence Loss Factor	0–0.2; blank by default
Two-Phase Loss Factor	0.8–1, starts at **empty**; no user correction is applied and the status says so
Ignition Delay (ms)	1–1000, starts at 50
Igniter Target Pressure (P/Pc)	0.02–0.5, starts at 0.15
Atmospheric Pressure (kPa)	1–200, starts at 101.325
Test Altitude (m)	0–30,000, starts at 0
Ambient Temperature (K)	180–340, starts at 298

C.3 Liquid Engines page

Field	Accepted range and starting value
Fuel Type	RP-1, LH2, Methane, MMH, UDMH, starts at RP-1
Oxidizer Type	LOX, N2O4, starts at LOX
Fuel / Oxidizer Density (kg/m^3)	20–2500; blank by default
Fuel / Oxidizer Boiling Point (K)	20–900 / 60–500, starts at 500 / 90.2
Fuel / Oxidizer Freezing Point (K)	10–400 / 40–350, starts at 200 / 54.4
Fuel / Oxidizer Viscosity (Pa·s)	10^{-6}–0.1; blank by default
Fuel Heat of Combustion (MJ/kg)	1–150, starts at 43.1
Fuel Heat Capacity C_p (J/kg·K)	500–20,000, starts at 2090
Thermal Conductivity λ_f (W/m·K)	0.01–5, starts at 0.145
Target Mixture Ratio (O/F)	0.5–20; blank by default
Stoichiometric O/F	starts at 3.4
O/F Range Minimum / Maximum	0.2–20, starts at 2.0 / 3.2
Combustion Efficiency η_c (%)	50–100, starts at 97
Engine Cycle Type	six entries, starts at Pressure-Fed
Target Thrust (N)	100–10,000,000; blank by default
Chamber Pressure P_c (bar)	10–500; blank by default
Feed System Pressure (bar)	1–1000, starts at 130
Turbopump Efficiency η_{tp} (%)	20–95, starts at 75
Generator Gas Temperature (K)	400–2500, starts at 1000

Field	Accepted range and starting value
Turbine Inlet Pressure (bar)	1–1000, starts at 150
Turbine Expansion Ratio	1.2–60, starts at 4
Feed Line Length (m)	0.05–50; blank by default
Feed Line Inner Diameter (mm)	0.5–500; blank by default
Feed Line Wall Thickness (mm)	0.1–50; blank by default
Valve Closure Time (ms)	0.1–600,000; blank by default
Tank Pressurization Gas	Auto, Helium, Nitrogen, Autogenous, starts at Auto
Injector Type	five entries, starts at Like-on-Like Impinging
Number of Injector Elements	1–20,000, starts at 100
Fuel / Oxidizer Injection Velocity (m/s)	1–150, starts at 25 / 40
Fuel / Oxidizer Orifice Diameter (mm)	0.1–25, starts at 2.0 / 2.5
Injector Pressure Drop ΔP (**bar**)	0.2–200, starts at 20; a value outside the band is not refused: it is ignored, the auto value (20% of P_c) is used, and the answer names the field
Discharge Coefficient C_d	0.2–1, starts at 0.7
Chamber Diameter D_c (mm)	10–5000 as declared; the effective limit depends on the contraction ratio band. Blank by default, derived from the contraction ratio
Characteristic Length L^* (m)	0.3–5, starts at 1.2
Contraction Ratio ε_c	1.5–60, starts at 4
Chamber Wall Thickness (mm)	0.2–150, starts at 5
Chamber Surface Roughness (μm Ra)	0.05–100, starts at 3.2
Cooling Type	five entries, starts at Regenerative (Fuel Cooling)

The chamber diameter band is declared, not enforced. A value outside 10–5000 mm does not refuse the request; it is ignored, the diameter is derived from the contraction ratio instead, and the answer names the field. A value *inside* the band is not safe either: if it pushes the contraction ratio outside its own band, it is dropped the same way and a second warning says so. On a chamber sized around a 33 mm throat, for example, 100 and 200 mm are applied as typed while 500 and 1000 mm are dropped back to the derived 75.7 mm. Treat the number you type here as a request, and read the published chamber diameter back before you trust it.

Field	Accepted range and starting value
Nozzle Expansion Ratio ε_e	1.5–300, starts at 50
Nozzle Type	Conical 15°, Bell 80%, Bell 60%, Aerospike, Dual-Bell, starts at Conical
Nozzle Throat Diameter (mm)	1–500, starts at 50
Cooling Channel Count	4–2000, starts at 80
Coolant Flow Rate (% of fuel flow)	5–100, starts at 100
Film Cooling (% of fuel flow)	0–30, starts at 0
Coolant Inlet Temperature (K)	20–700, starts at 300
Maximum Wall Temperature (K)	300–3000, starts at 800
Engine Start / Shutdown Time (s)	0.01–30, starts at 2.0 / 1.5
Minimum Throttle (% of max thrust)	5–100, starts at 40
Throttle Response Time (s)	0.01–10, starts at 0.5
Chill-Down Time (s)	0–3600, starts at 30
Maximum Burn Duration (s)	0.1–100,000, starts at 400
Target Thrust-to-Weight T/W	1–500, starts at 60
Structural Safety Factor	1.1–10, starts at 2.5
Gimbal Range (± degrees)	0–45, starts at 8
Actuator Response Time (ms)	1–1000, starts at 50
Acoustic Environment (dB)	80–200, starts at 165
Min / Max Ambient Temperature (K)	3–500, starts at 200 / 350

C.4 Canonical envelopes

Section 11 lists the per-endpoint bands that a rejection quotes. Those are derived from the canonical envelope, the repository-wide statement of each quantity in SI units. The length envelopes below are the ones worth knowing, because they are wide enough that a unit error of a factor of 1000 can still land inside one.

Quantity	Canonical envelope	Form view
Chamber diameter	0.01–5.0 m	solid form 10–5000 mm
Chamber length	0.05–20.0 m	
Grain length	0.05–10.0 m	solid form 50–10,000 mm
Throat diameter	0.001–2.0 m	solid form 1–500 mm
Exit diameter	0.002–5.0 m	solid form 1.5–2500 mm
Port diameter	0.005–1.0 m	
Web thickness	0.001–0.5 m	
Tank pressure	5–700 bar	
Burn time	0.1–300 s	hybrid form 0.1–1000 s
Thrust	10 N–1 MN	hybrid 1–10^7 N, liquid 100–10^7 N

C.5 Bands the checks apply after the calculation

These do not refuse a request. They decide whether a warning is raised, and the warning prints the band.

Check	Band
Specific impulse, c^* and O/F	Per propellant pair; the table in Section 6 lists all three. A specific impulse below 80% or above 120% of the upper edge is critical, as is a c^* below 90% of its lower edge
Expansion ratio	4–250
Initial port diameter	5–100 mm
Final port diameter	No band of its own. It is judged twice, both rows below: through the growth ratio, and against the chamber diameter
Port growth ratio $d_{\text{final}}/d_{\text{initial}}$	1.2–3.0; below the lower edge a warning, above the upper edge critical
Final port diameter against the chamber	critical above 80% of the chamber diameter
Injector $\Delta P_{\text{inj}}/P_c$	0.15–0.35; 0.20 recommended. The lower edge is a chug stability criterion, the upper edge a feed-system cost criterion
Injector orifice Reynolds number and exit velocity	1000–200,000; 10–100 m/s. The orifice hole diameter is *not* judged here; it is gated on the form instead (liquid page, 0.1–25 mm)
Oxidizer port mass flux	warning at 600 kg/m^2·s, the lower edge of the 600–700 flooding band
Grain and chamber slenderness L/D	warning above 10
c^* efficiency	applied as given; 0.5–1.05 is the physically defensible band, and measured hybrids are typically 0.90–0.97. Note that the Uncertainty panel's input table quotes a different literature band for the same quantity (0.85–0.95 for hybrids, 0.92–0.99 for liquids), because the two came from different passages of the same sources
Summerfield separation factor k	accepted 0.35–0.40, default 0.40

Chamber pressure is not on this list, and that is not an omission. No post-calculation warning is raised on chamber pressure at all. Its real gates sit on the form side and are refusals, not warnings: the solid endpoint accepts 5–200 bar, the liquid endpoint 10–500 bar, and the hybrid quick-geometry and find-optimum endpoints 1–500 bar, each refusing an out-of-band value with HTTP 400 and naming the band. The hybrid calculation form has no chamber-pressure band of its own; the "at least 1" floor you meet on screen is the input widget's own minimum. If you

press *Calculate* at 500 bar you will still see a warning, but read it carefully: it comes from the thrust-coefficient identity $C_F \equiv F/(A_t P_c)$, which is a consistency check between three numbers you supplied, not a statement that 500 bar is out of range.